

ONT 16S Full-Length - Bacteria

Bacterial 16S Full-Length

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Glossary

Short definitions for common terms used in this report.

Term	Definition
99% OTU	Operational taxonomic unit clustered at 99% sequence identity from full-length 16S representative sequences.
EMU	A full-length 16S read-level taxonomic profiling method used for family, genus, and species composition charts.
Raw reads	Sequencer reads before primer trimming, length filtering, quality filtering, clustering, or taxonomy filtering.
Post-filter reads	Reads retained after primer trimming and length/quality filtering before final feature-table filtering.
Final feature-table reads	Reads represented in the final bacterial feature table after clustering, chimera screening, and report-level filtering.
Rarefaction sampling depth	The fixed read depth used to compare diversity across samples on the same scale.
Relative abundance	The fraction of a sample assigned to a taxon after normalizing that sample to 1.00.
CPM	Counts per million; a per-sample normalization that rescales predicted functional abundance to the same total before plotting.
Heatmap normalization	Taxonomy heatmaps use per-sample relative abundance. Functional heatmaps use per-sample CPM normalization, $\log_{10}(\text{CPM} + 1)$, then row scaling within each function so colors show relative enrichment/depletion patterns rather than absolute microbial load or sequencing depth.
Alpha diversity	Within-sample diversity, such as observed 99% OTUs, Shannon, Simpson, and Faith PD.
Beta diversity	Between-sample community difference, often visualized by PCoA plots.
PCoA	Principal coordinates analysis; a two-dimensional view of sample distances.
PERMANOVA	A group-level test for differences in community composition based on a distance matrix.
PERMDISP	A test of whether group dispersion differs; useful context for PERMANOVA results.
ALDEx2	A compositional differential-abundance method used to screen taxa or functions associated with groups.
Faith PD	A phylogenetic alpha-diversity metric based on branch length represented in a sample.
NSTI	Nearest sequenced taxon index; lower values generally indicate stronger PICRUSt2 prediction support.

1 - Methods

ONT long-read amplicon data targeting the approximately 1.5 kb full-length bacterial 16S rRNA gene were processed using forward primer AGAGTTTGATCMTGGCTCAG and reverse primer TACGGYTACCTTGTTACGACTT. Reads were quality-profiled with NanoPlot v1.46.2, primer-trimmed with Cutadapt v5.1, and filtered with chopper v0.11.0 to retain reads passing minimum length 1300 bp, maximum length 1700 bp, minimum Q-score 15. Filtered reads were screened for chimeras with VSEARCH v2.22.1_linux_x86_64 using full-length dereplication and de novo chimera detection with uchime_denovo. De novo feature inference was performed with FOI NanoPulse; BLAST-based and Kraken2-based taxonomic pre-assignment were disabled. Per-sample NanoPulse cluster representatives were pooled and clustered at 99% identity with VSEARCH to generate shared 99% clustered features/OTUs, and chimeric cohort representatives were removed with VSEARCH uchime_denovo. This assay is designed to detect bacteria using full-length 16S primers, but archaeal assignments may appear in the results when classified by the database. Features classified under domain Bacteria or Archaea were retained for feature-level taxonomy, diversity, phylogeny, and PICRUST2 analysis; domain-unclassified, mitochondrial, chloroplast, eukaryotic, and other configured off-target features were removed. Read-level taxonomic composition was summarized with EMU v3.6.2 and rolled up to family, genus, and species levels. The EMU reportable figure/taxonomy tables use the configured report filters (minimum relative abundance 0.005 and minimum estimated reads 10 in at least one sample). Taxonomic ALDEx2 comparisons use the unfiltered EMU estimated-count matrices aggregated by rank, rounded to non-negative integers with zeros retained before pairwise testing (mc.samples=128, denom=all, Welch-test FDR correction within each contrast). QIIME 2 v2026.1.0 with the SILVA classifier database (138.2; classifier artifact: silva-138-2-full-length-classifier.qza) was retained for 99% clustered feature taxonomy, representative-sequence phylogeny, and phylogenetic diversity outputs. Alpha diversity was assessed using Shannon, Simpson, and Faith phylogenetic diversity metrics after rarefaction to 800 reads per sample, with Kruskal-Wallis group testing when supported by the metadata. Beta diversity was assessed using Bray-Curtis, unweighted UniFrac, and weighted UniFrac distances with PCoA ordination, PERMANOVA, and PERMDISP where group sizes supported testing. PICRUST2 v2.6.3 was used to predict community-level functional potential from a singleton-filtered, 97% clustered representative-sequence input, with EC, KO, and MetaCyc pathway outputs retained with NSTI tracking. Differential taxonomic and functional abundance was assessed using ALDEx2 v1.42.0 in R.

Table 1. Sequencing provenance recorded for this analysis.

Field	Value
Flow cell / chemistry	R10.4.1
Library kit	Ligation Sequencing Kit V14 (SQK-LSK114)

Table 2. Reference database and EMU filtering settings used for this analysis.

Database/setting	Reported value
SILVA classifier	138.2
Classifier artifact	silva-138-2-full-length-classifier.qza
EMU database	default EMU database
EMU database version/build	v3.6.2
EMU database date	September 17, 2020
EMU minimum abundance	0.0001
EMU reportable minimum relative abundance	0.005
EMU reportable minimum estimated reads	10

Table 3. NanoPulse clustering input-size policy. This cap is applied before de novo clustering and is separate from rarefaction depth and final feature-table depth.

Field	Value
Policy stage	NanoPulse clustering/UMAP input-size selection
Purpose	Limit per-sample clustering input size for comparable and tractable de novo clustering
Selected tier	low
Selected UMAP set size	5000
Cohort minimum post-filter reads	1987
Subsampled samples	LPW47, LPW46, CPW44, CPW43

Workflow Overview



2 - Analytical Summary

This exploratory analysis report summarizes 9 biological samples from an ONT full-length 16S bacteria-focused assay. The automatically selected rarefaction depth was 800 reads per sample; lowest final sample depth 837 (APW61).

- Key signal: rarefaction adequacy was 9/9 samples; alpha-diversity tests were not statistically significant when $p \geq 0.05$ (Observed 99% OTUs $p=0.737$, Shannon $p=0.957$, Simpson $p=0.957$, Faith PD $p=0.561$).
- Key signal: 99% OTU Bray-Curtis was non-informative because all pairwise distances were 1.000; interpret beta diversity mainly from EMU species-level Bray-Curtis and UniFrac metrics (Bray-Curtis $p=1.000$, unweighted UniFrac $p=0.434$, weighted UniFrac $p=0.096$).
- Risk: weighted UniFrac PERMDISP was significant ($p=0.030$), so weighted UniFrac PERMANOVA should be interpreted cautiously because group dispersion differs.
- Key signal: differential testing found 0 significant taxonomic features and 0 significant predicted functional features at the configured FDR threshold.
- Risk: extraction blanks, PCR negative controls, and positive controls were not supplied or not identified in metadata, so process-control performance could not be assessed for this run.
- Action: interpret group-level findings as exploratory unless supported by larger cohorts, appropriate controls, and targeted validation.

3 - Sample Details

Table 4. This table summarizes the final bacterial analysis depth for each biological sample. Group is the comparison group used in the report, Sample Name is the user-facing metadata label, and Total Reads is the sample depth in the 99% clustered feature/OTU table after quality filtering, feature inference, and off-target removal.

Group	Sample Name	Total Reads
Control	CPW42	1137
Control	CPW43	2809
Control	CPW44	2905
Latent	LPW46	2596
Latent	LPW47	3858
Latent	LPW48	2619
Active	APW59	2326
Active	APW60	1150
Active	APW61	837

4 - Taxonomic Composition

Taxonomic composition is summarized from EMU full-length 16S read-level profiling and rolled up to family, genus, and species levels for the main charts. Each sample is normalized to a total abundance of 1.00 prior to plotting. Rank-specific figures exclude unclassified or unresolved labels at that rank, so displayed family/genus/species panels may sum below 100% when part of a sample lacks that rank assignment.

Relative abundance values are compositional and do not represent absolute microbial load, cell counts, or organism concentration. 16S taxonomic assignments are research-level estimates and do not confirm infection, pathogenicity, or clinical diagnosis; confirm relevant organisms using targeted testing, culture, or validated diagnostic methods.

Top 20 family taxa grouped by metadata group

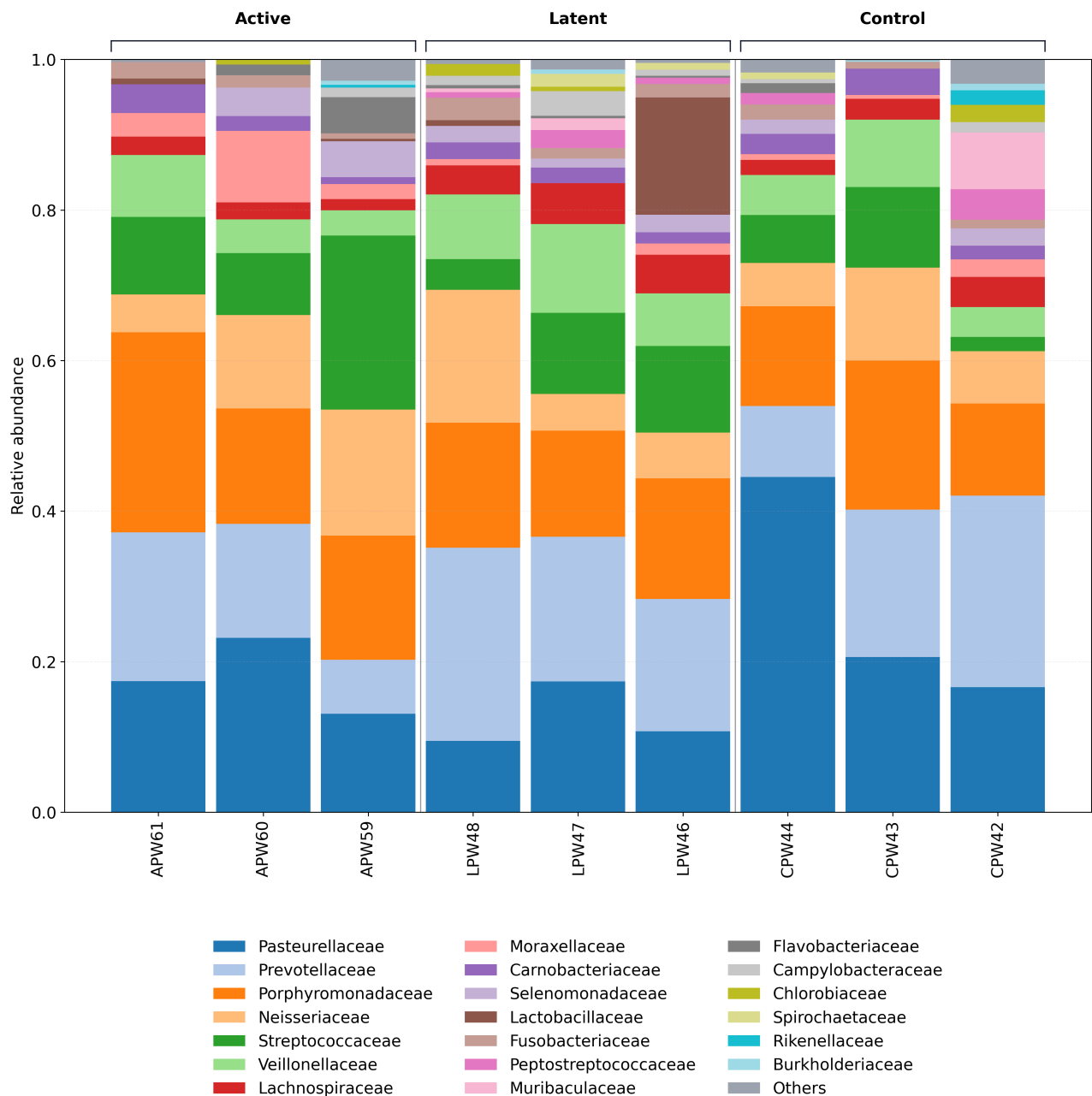


Figure 1a. EMU-derived family-level stacked bar chart of relative abundance by sample, ordered and separated by metadata group. Each bar is one sample and each color band is a family-level taxon. Relative abundance is normalized per sample (sum = 1.00), so the figure compares composition rather than absolute read count.

Top 20 genus taxa grouped by metadata group

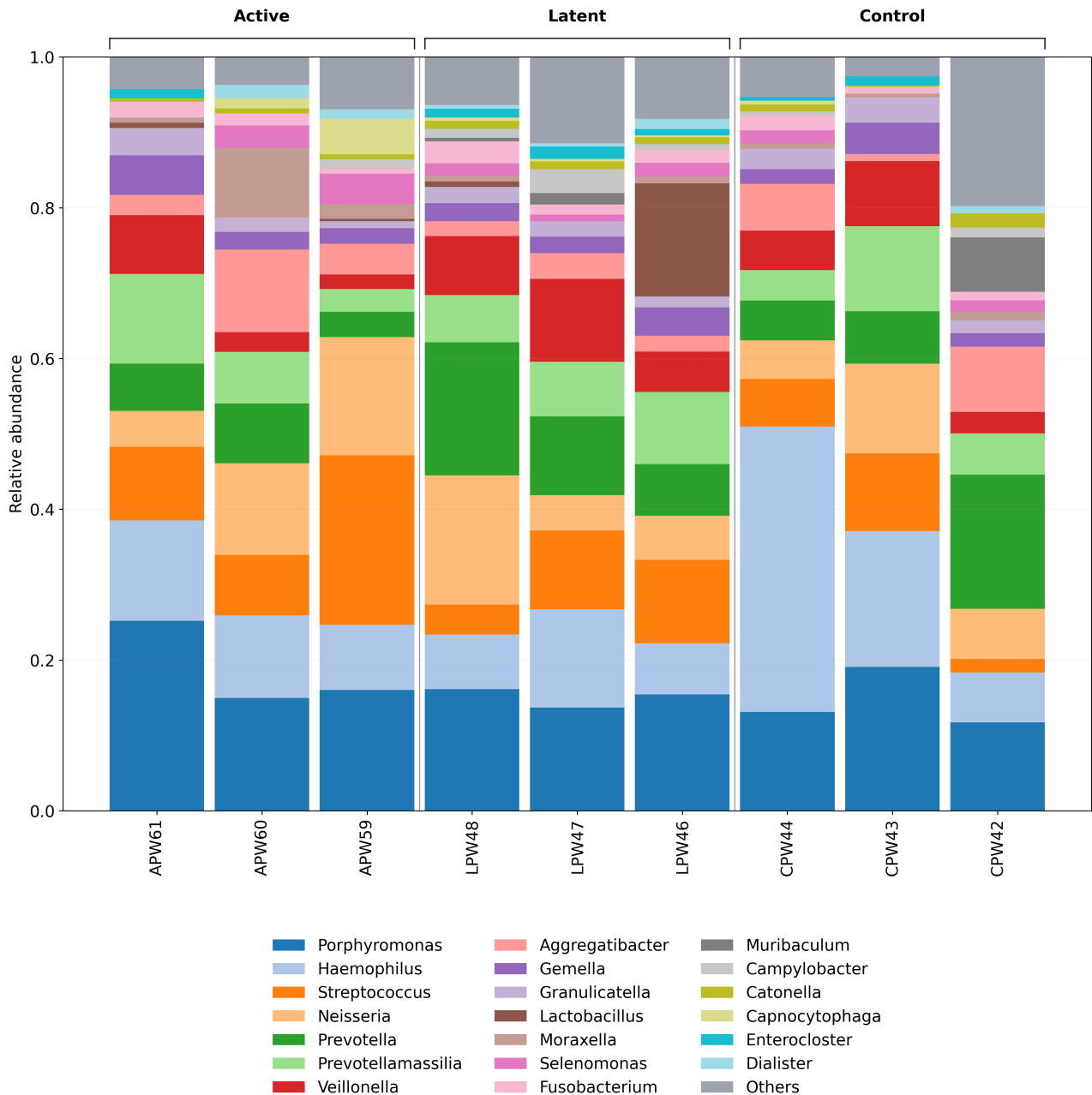


Figure 1b. EMU-derived genus-level stacked bar chart of relative abundance by sample, ordered and separated by metadata group. This view helps identify dominant genera and abrupt composition shifts between groups. Relative abundance is normalized per sample (sum = 1.00), so the figure compares composition rather than absolute read count.

Top 20 species taxa grouped by metadata group

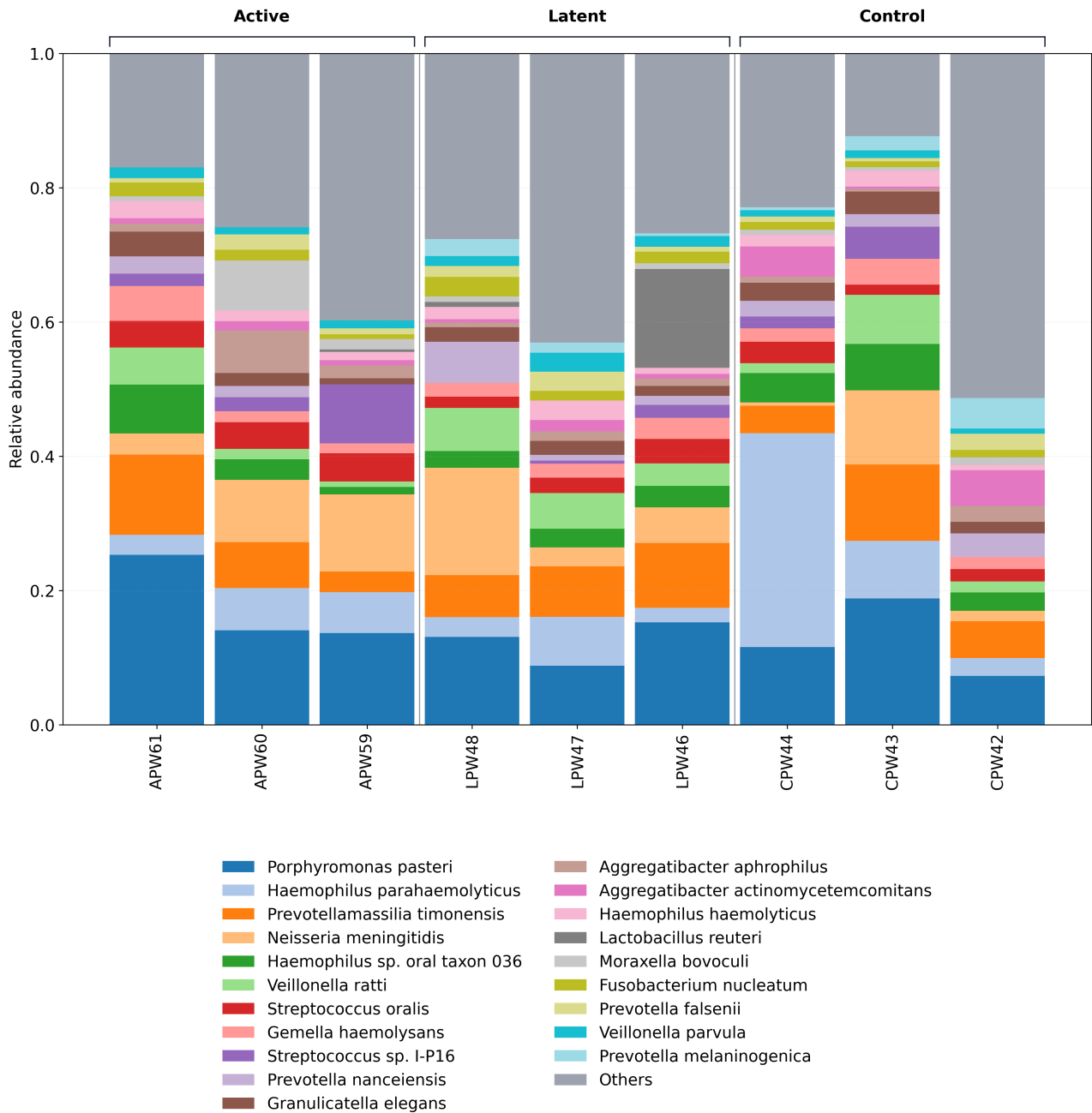


Figure 1c. EMU species-level stacked bar chart of relative abundance by sample, ordered and separated by metadata group. Species names come from EMU full-length 16S microbial profiling. Relative abundance is normalized per sample (sum = 1.00), so the figure compares composition rather than absolute read count.

Top 20 family taxa group mean

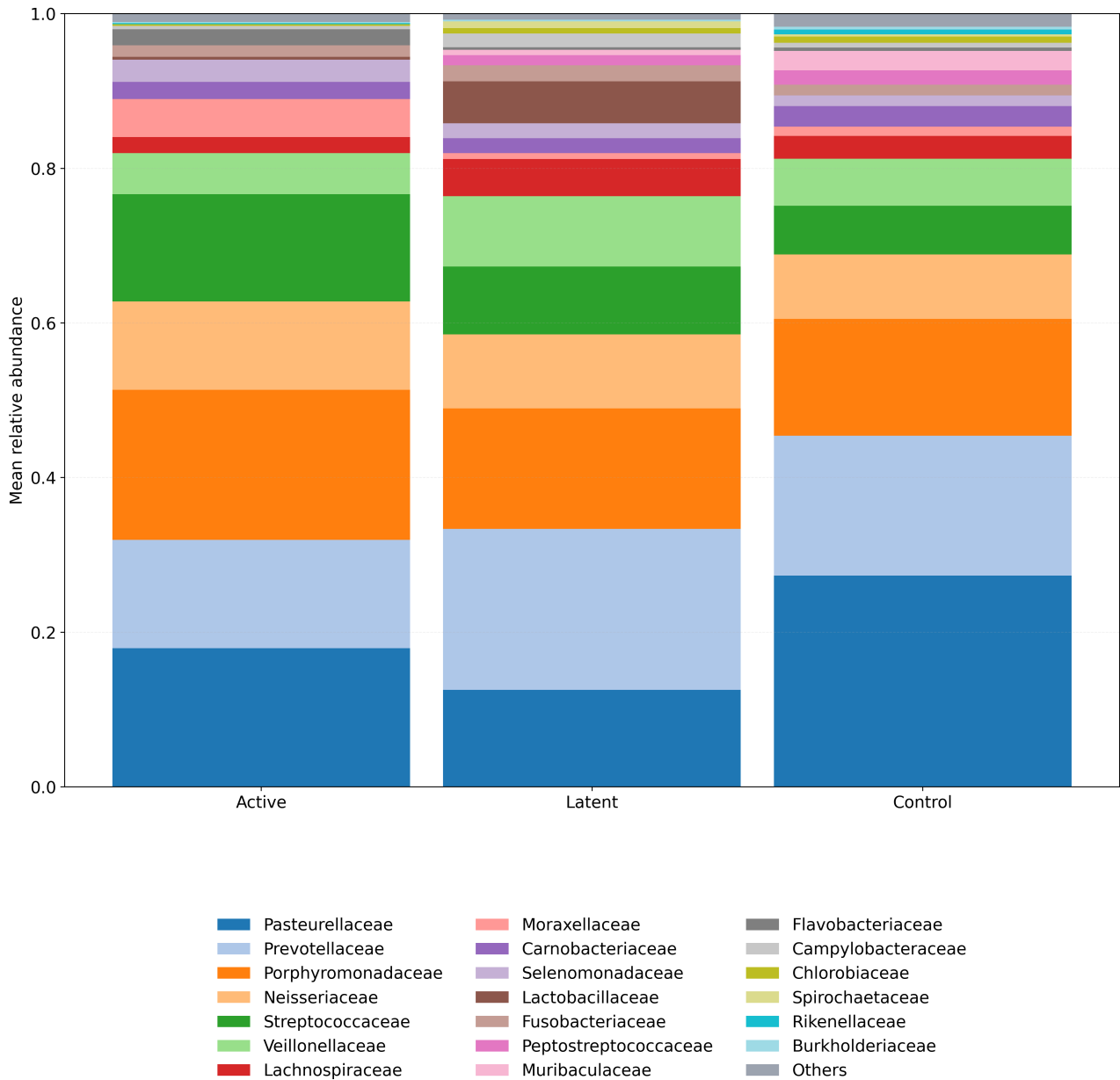


Figure 1d. EMU family-level stacked bar chart of group mean relative abundance. Each bar is the mean composition for one metadata group after per-sample normalization.

Top 20 genus taxa group mean

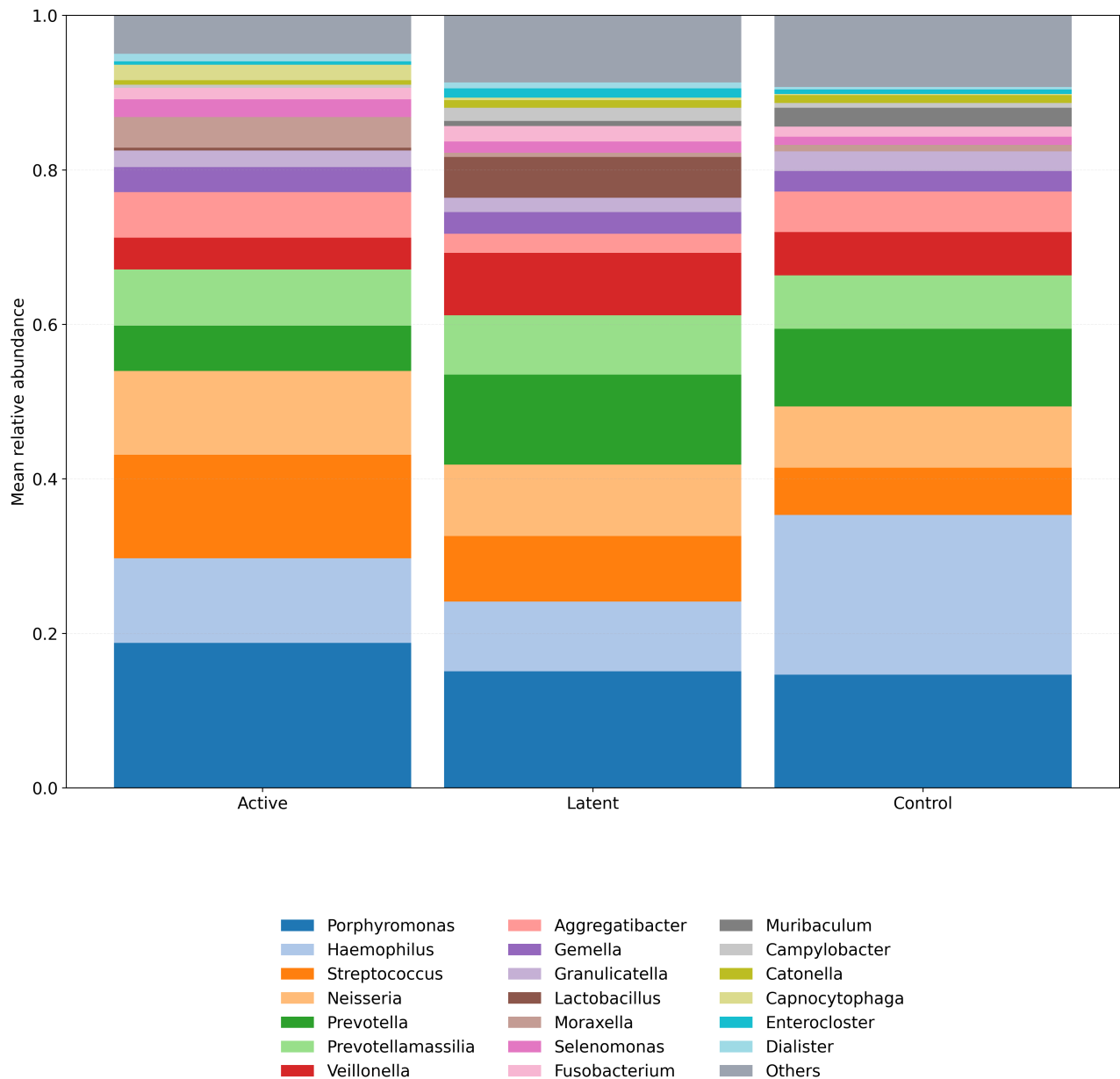


Figure 1e. EMU genus-level stacked bar chart of group mean relative abundance. Each bar is the mean composition for one metadata group after per-sample normalization.

Top 20 species taxa group mean

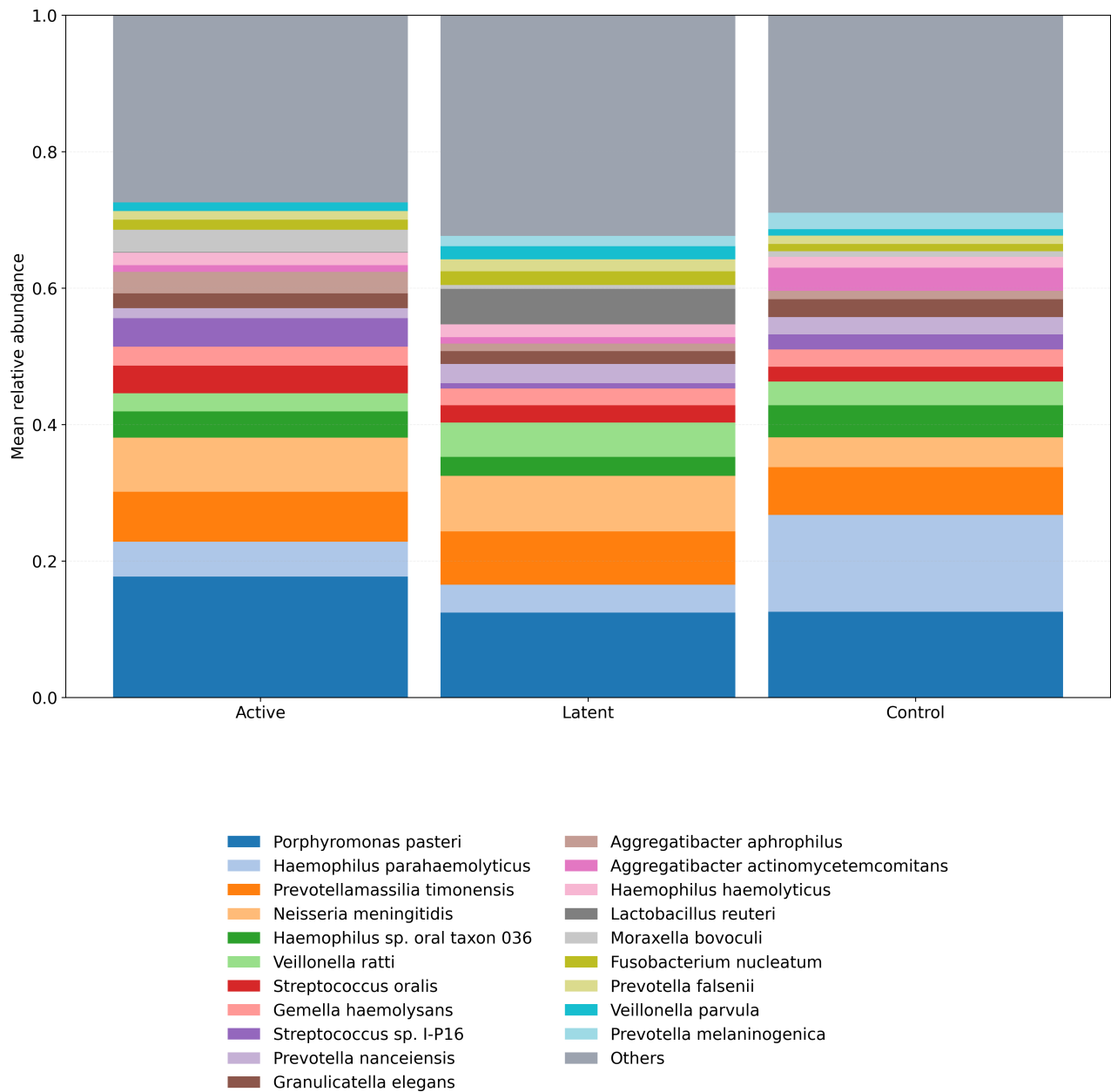


Figure 1f. EMU species-level stacked bar chart of group mean relative abundance. Each bar is the mean composition for one metadata group after per-sample normalization.

Top 28 family heatmap

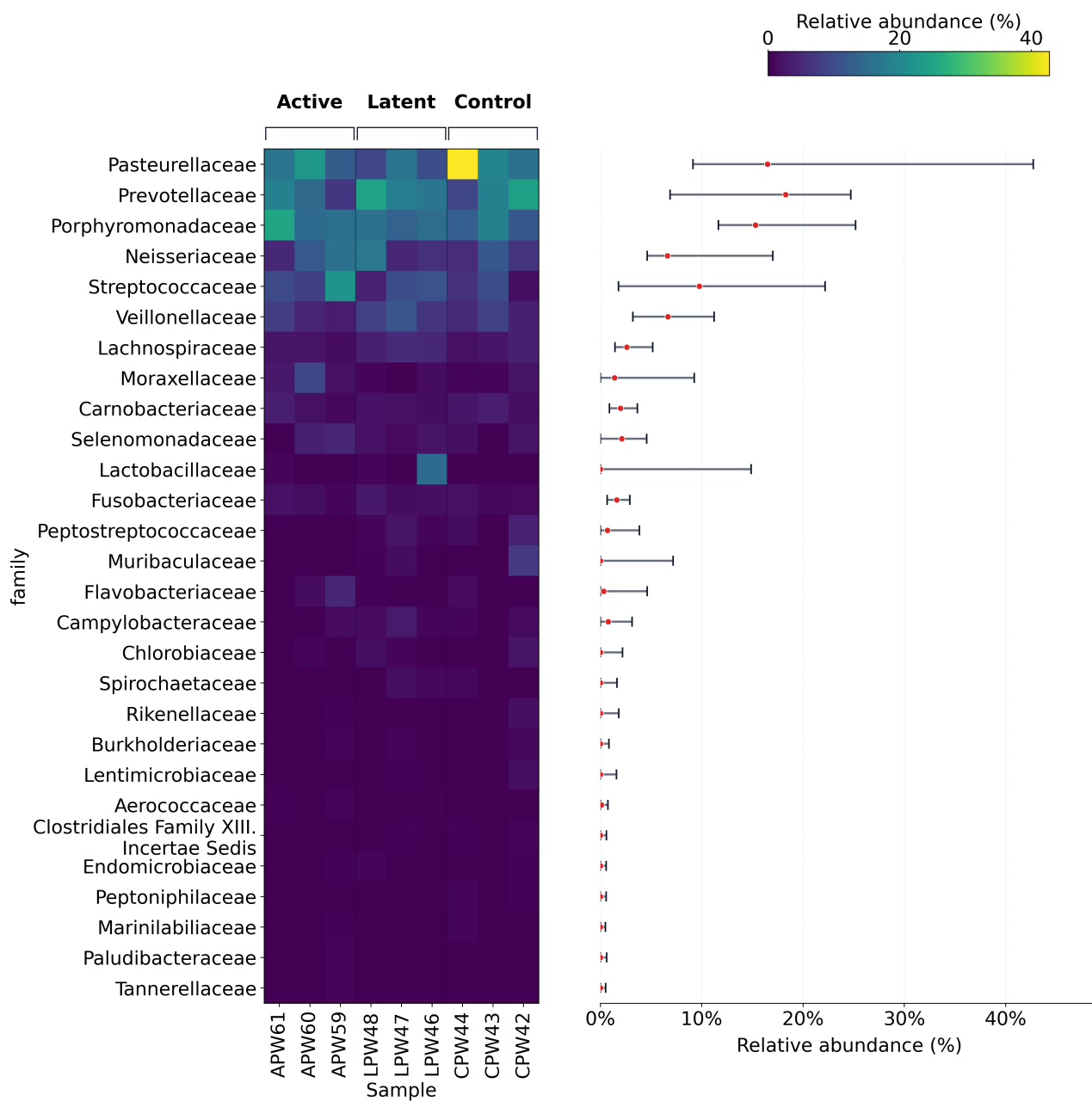


Figure 2a. EMU-derived family-level heatmap with samples on the x-axis and taxa on the y-axis. Color intensity encodes per-sample normalized relative abundance; yellow cells indicate stronger signal, while dark blue/purple cells indicate lower signal. The side chart shows each taxon's relative-abundance range across samples: grey line = min-max, black ticks = min/max, red point = median.

Top 30 genus heatmap

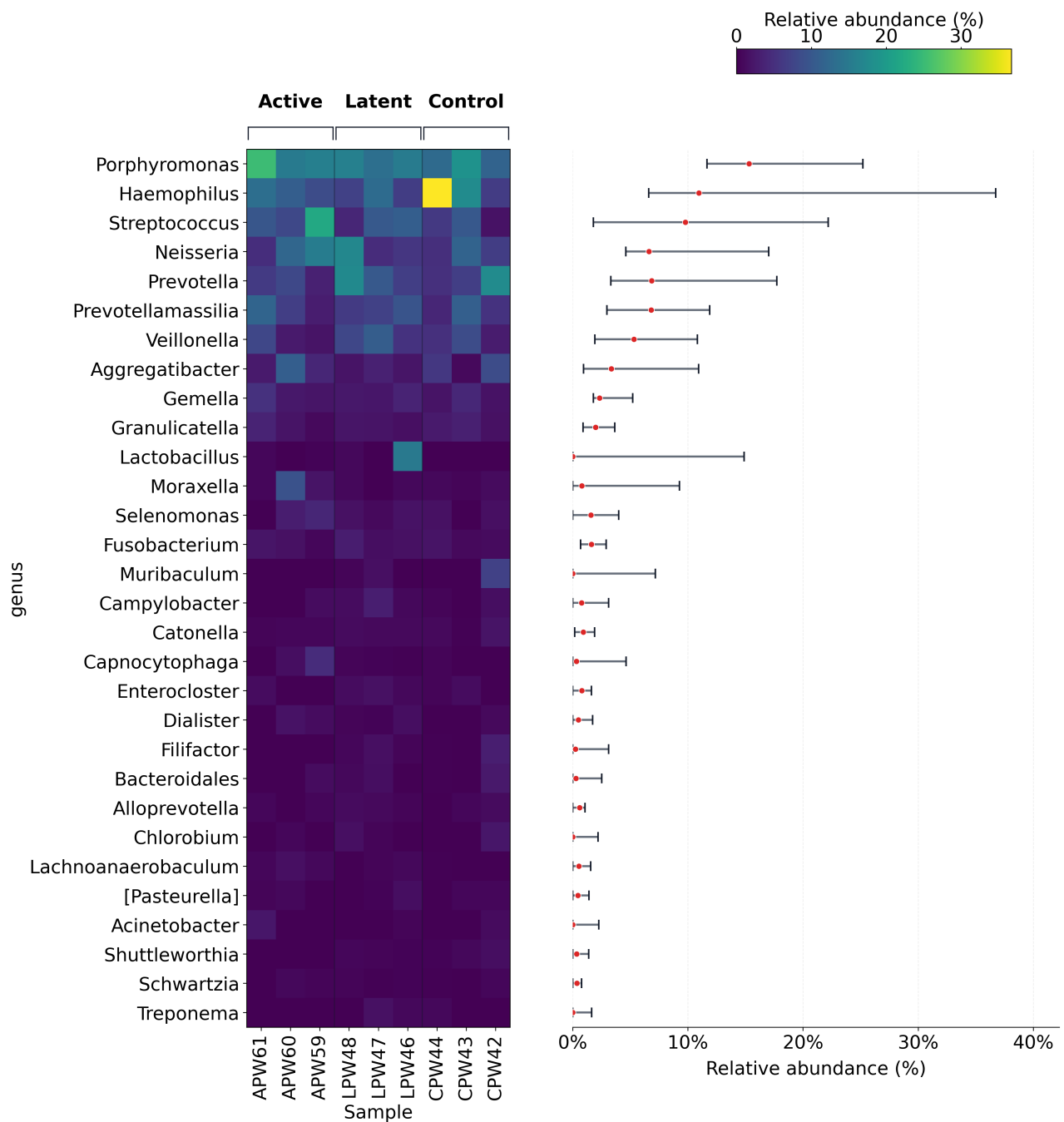


Figure 2b. EMU-derived genus-level heatmap with samples on the x-axis and taxa on the y-axis. Color intensity encodes per-sample normalized relative abundance; yellow cells indicate stronger signal, while dark blue/purple cells indicate lower signal. The side chart shows each taxon's relative-abundance range across samples: grey line = min-max, black ticks = min/max, red point = median.

Top 30 species heatmap

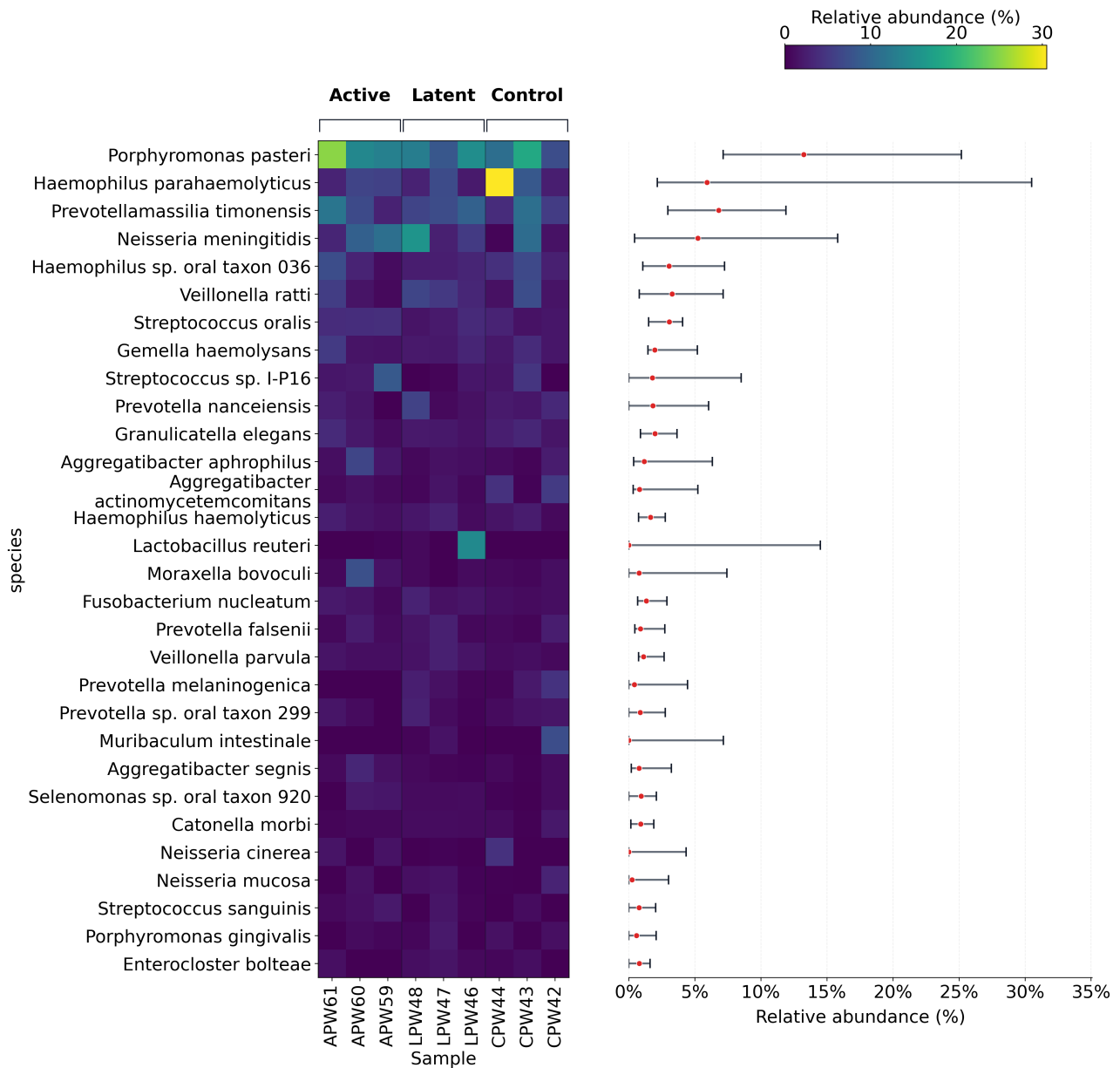


Figure 2c. EMU species-level heatmap with samples on the x-axis and species on the y-axis. Color intensity encodes per-sample normalized relative abundance; yellow cells indicate stronger signal, while dark blue/purple cells indicate lower signal. The side chart shows each taxon's relative-abundance range across samples: grey line = min-max, black ticks = min/max, red point = median.

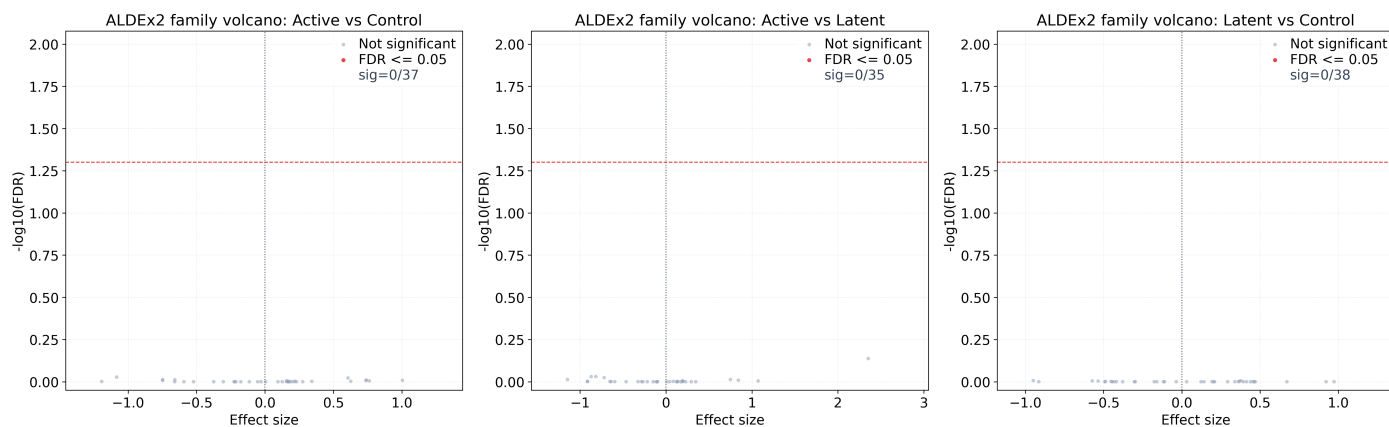


Figure 3. Family-level ALDEx2 differential-abundance volcano plots from EMU estimated-count tables rounded to non-negative integers. Each panel is one pairwise contrast; points farther from zero effect and above the FDR threshold have stronger evidence of group-associated taxonomic shifts.

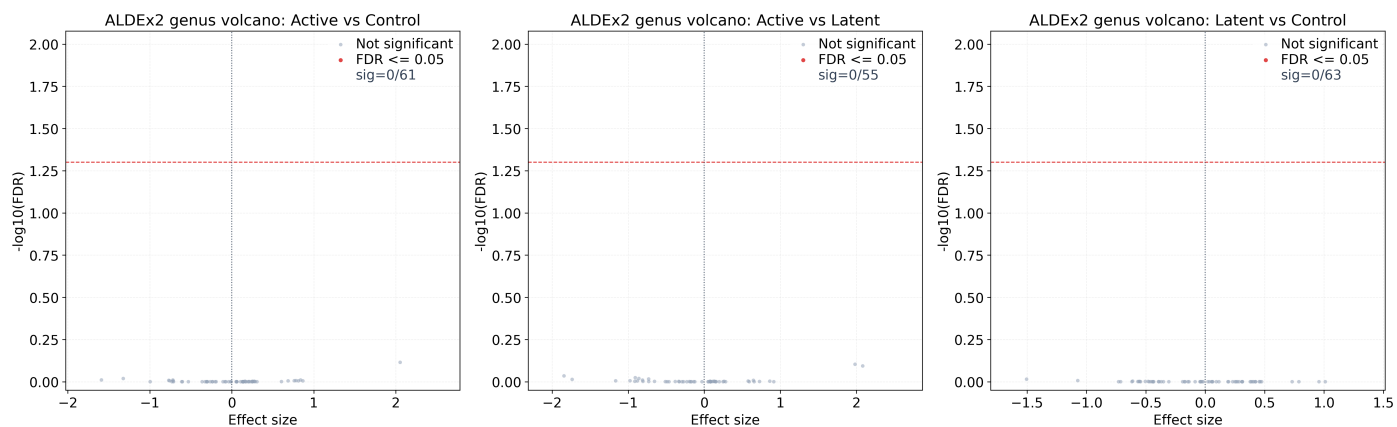


Figure 4. Genus-level ALDEx2 differential-abundance volcano plots from EMU estimated-count tables rounded to non-negative integers. Each panel is one pairwise contrast; points farther from zero effect and above the FDR threshold have stronger evidence of group-associated taxonomic shifts.

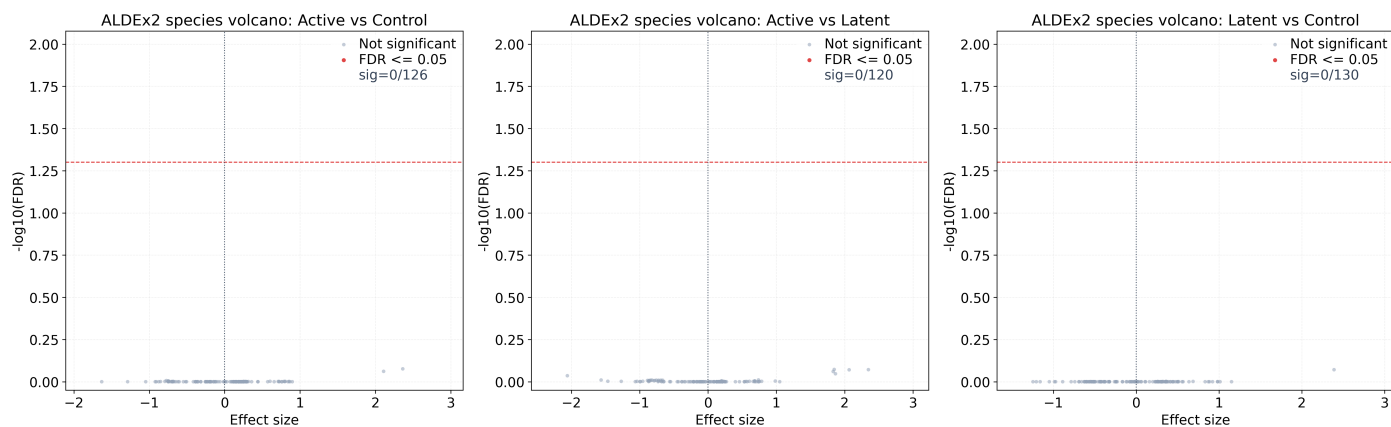


Figure 5. Species-level ALDEx2 differential-abundance volcano plots from EMU estimated-count tables rounded to non-negative integers. Each panel is one pairwise contrast; species-level 16S calls should be interpreted more cautiously than family/genus trends unless supported by stable replicate patterns.

- Key signal: when the same dominant taxa patterns appear across multiple samples in a group, group-level ecological interpretation is stronger.
- Risk: EMU provides species-level microbial profiles from full-length 16S reads, but closely related species can share highly similar 16S sequences, reference databases may be incomplete, and low-abundance species should be interpreted cautiously.
- Action: treat family/genus trends as primary evidence, and only keep species-level claims when they are stable across replicates and externally validated.

5 - Diversity and Community Structure

Alpha-diversity

Table 5. This table reports within-sample diversity. Group is the comparison group, Sample Name is the metadata label, Observed 99% OTUs is the number of detected 99% clustered features, Shannon Index increases with richness and evenness, Simpson Index increases when the community is more evenly distributed rather than dominated by a few taxa, and Faith PD adds phylogenetic alpha diversity using branch length from the rooted representative-sequence tree.

Group	Sample Name	Observed 99% OTUs	Shannon Index	Simpson Index	Faith PD
Active	APW61	13	2.05	0.82	1.67
Active	APW60	18	2.53	0.90	2.67
Active	APW59	30	2.93	0.92	3.81
Latent	LPW48	23	2.48	0.87	3.35
Latent	LPW47	23	2.64	0.91	4.30
Latent	LPW46	22	2.27	0.83	2.70
Control	CPW44	34	3.07	0.93	3.64
Control	CPW43	11	1.86	0.82	1.95
Control	CPW42	18	2.45	0.87	2.90

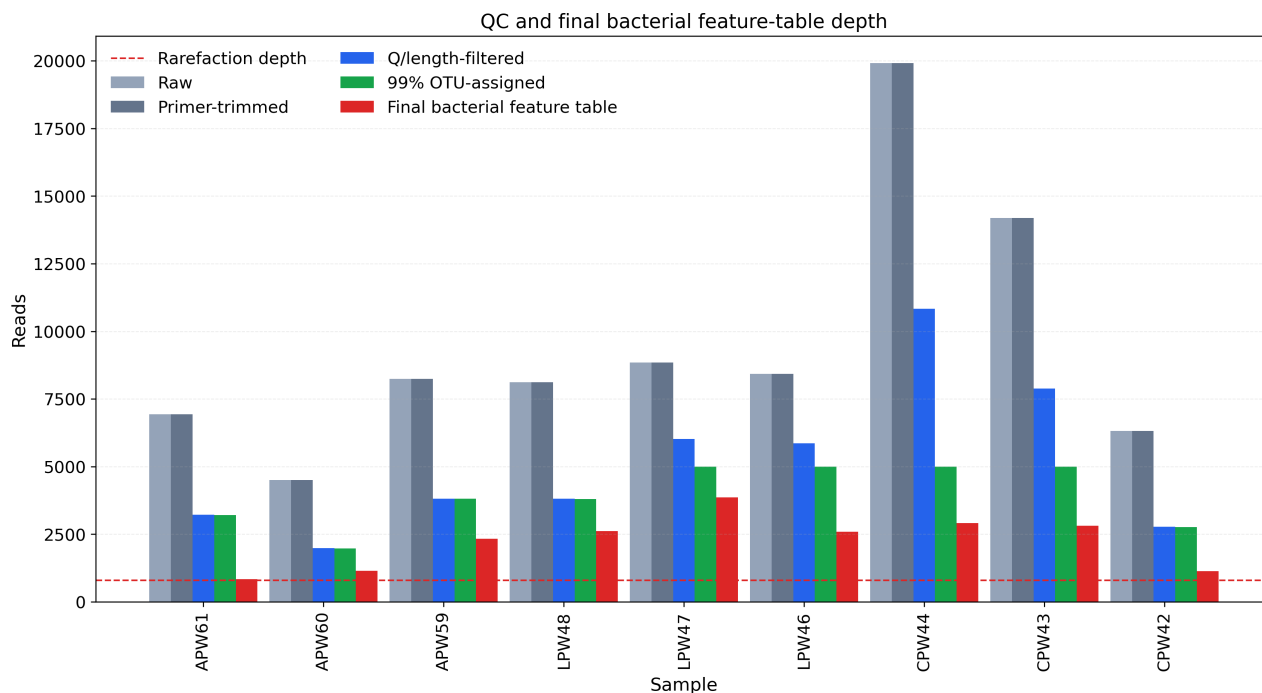


Figure 6. QC and final bacterial feature-table depth by sample. Grouped bars show raw, primer-trimmed, Q/length-filtered, 99% OTU-assigned, and final feature-table reads. The dashed line marks the selected rarefaction depth applied to final feature-table reads.

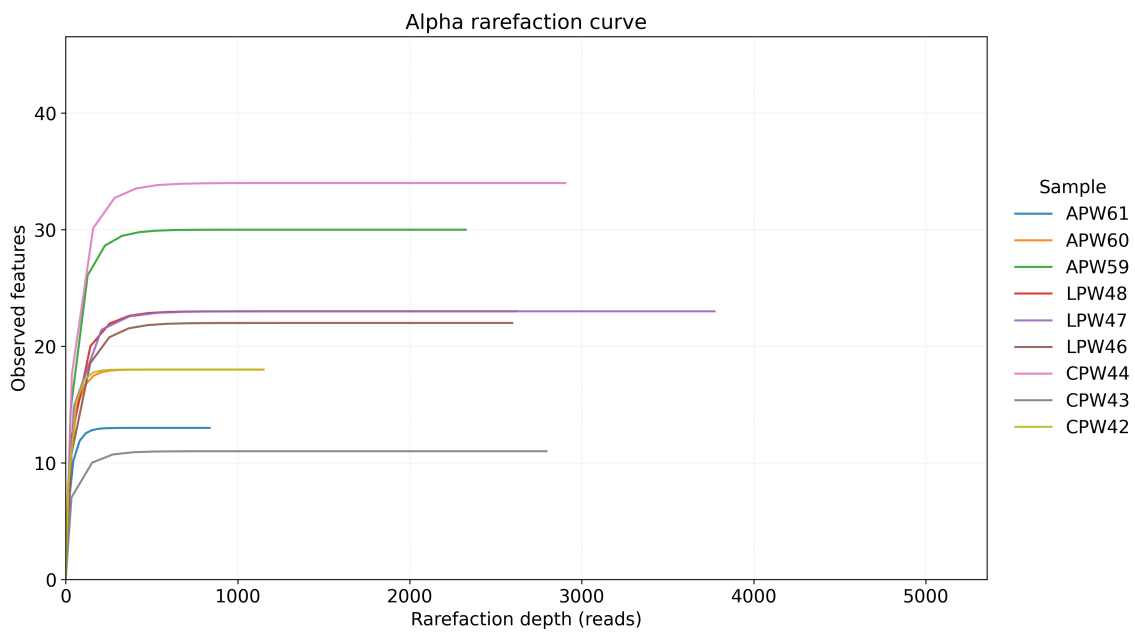


Figure 7. Alpha rarefaction curve (observed 99% OTUs vs read depth). Curves that flatten before the available depth suggest feature saturation, while curves that continue rising suggest additional depth may recover more features.

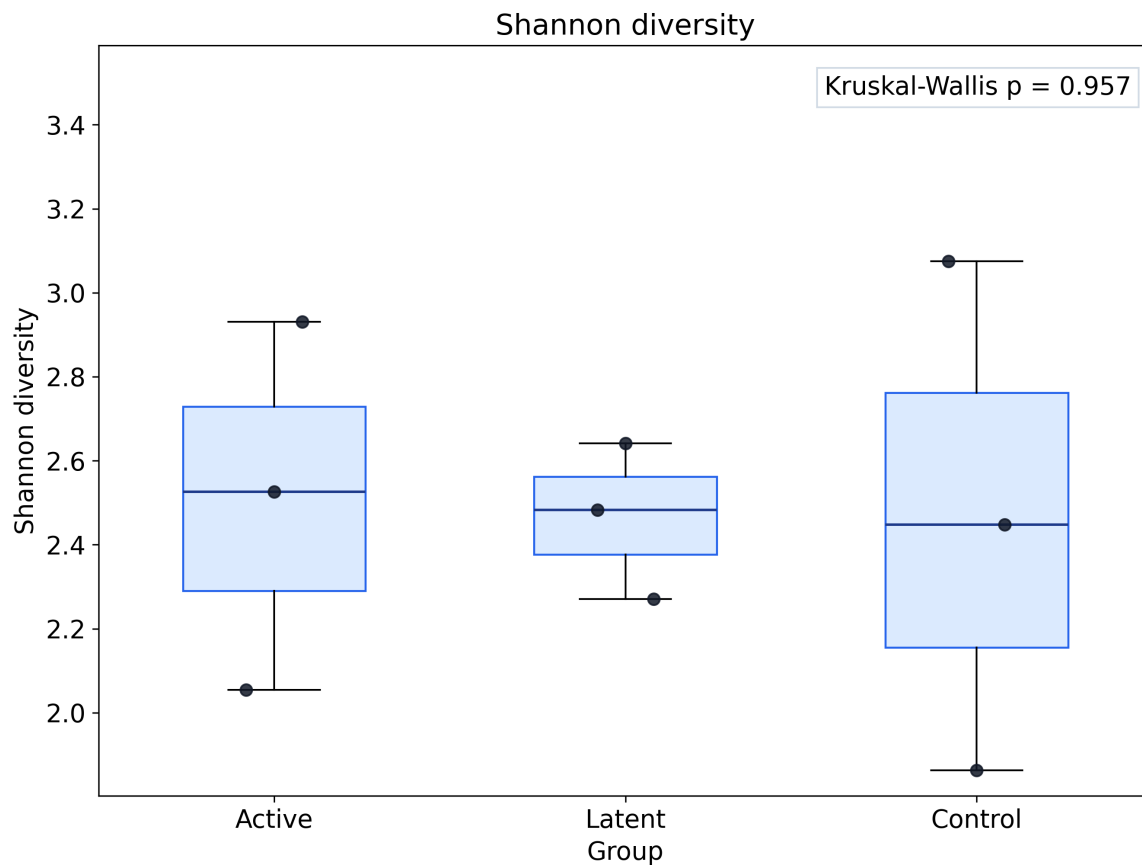


Figure 8. Kruskal-Wallis Shannon-diversity comparison across groups ($p=0.957$). This figure shows group-wise Shannon distributions with median and spread, and provides the formal between-group test context before interpretation.

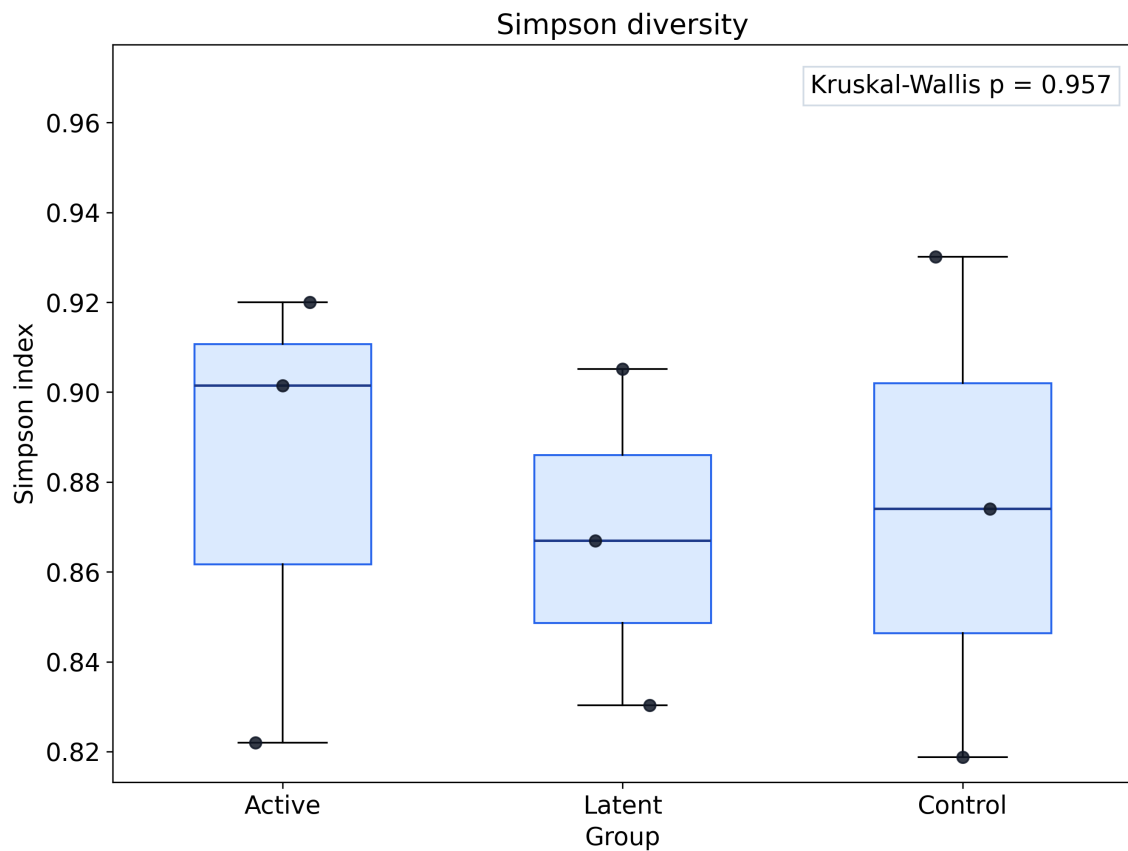


Figure 9. Simpson-diversity comparison across groups ($p=0.957$). Higher Simpson values indicate a more even community with less dominance by a few taxa, and complement Shannon-based alpha-diversity interpretation.

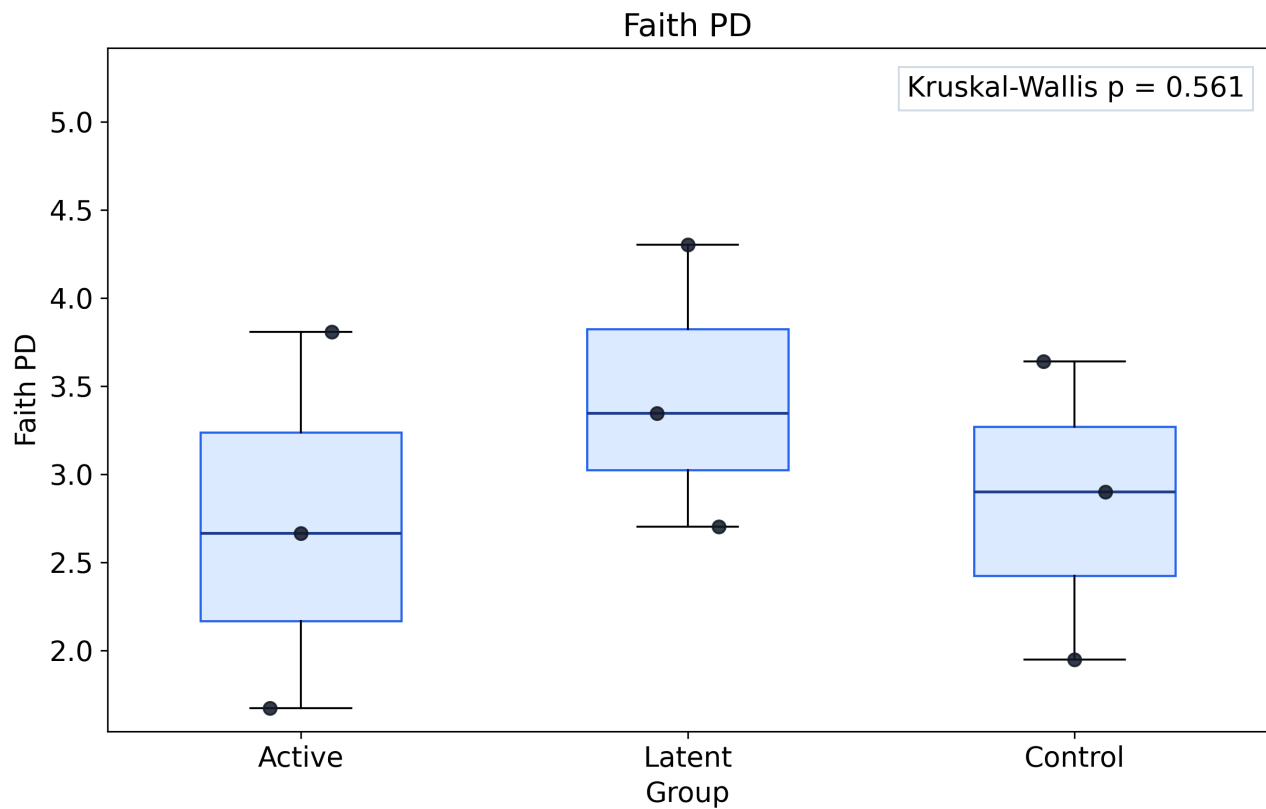


Figure 10. Faith phylogenetic diversity comparison across groups ($p=0.561$). Higher values indicate broader phylogenetic breadth in the detected community.

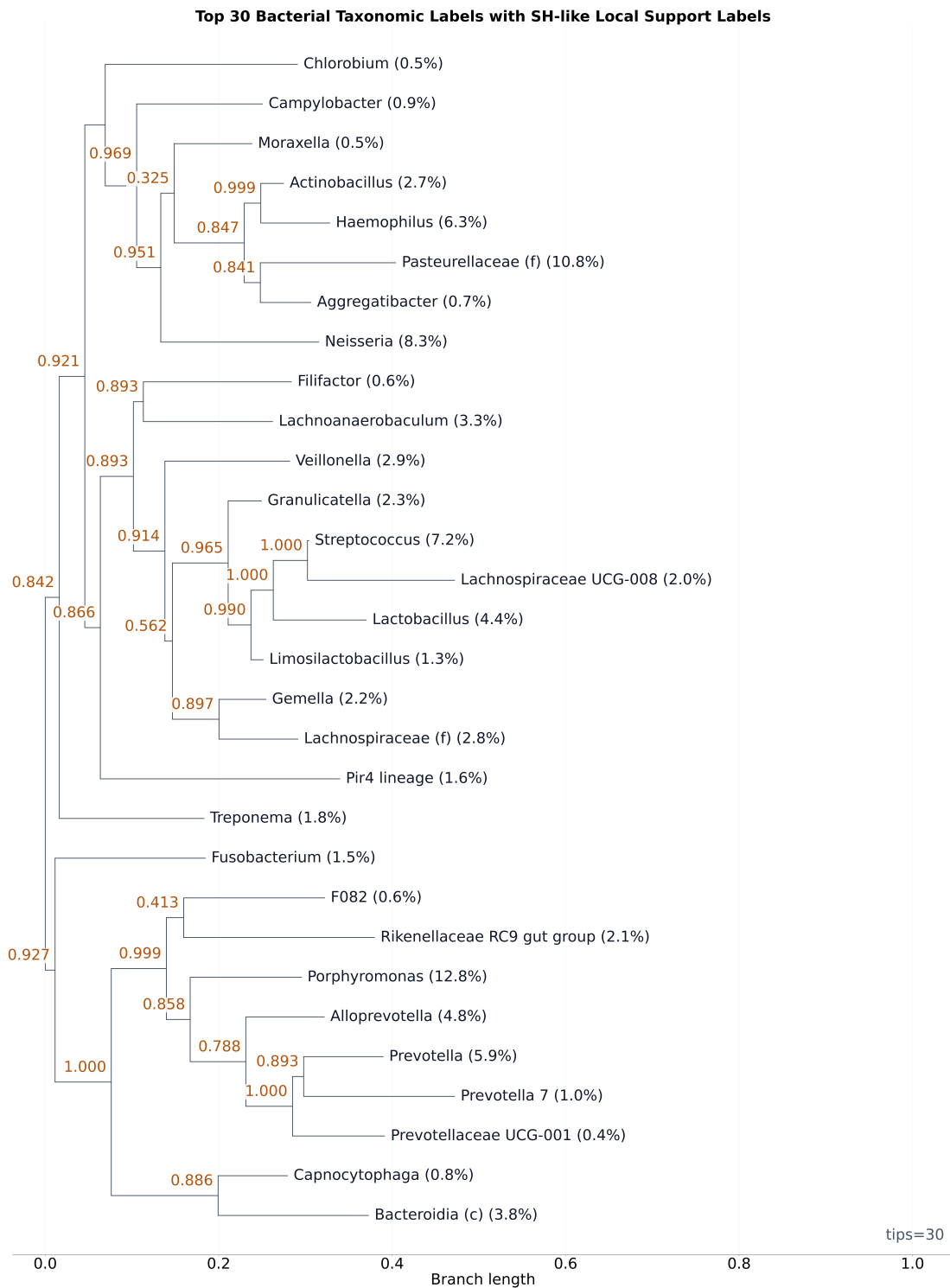


Figure 11. Top 30 bacterial taxonomic labels phylogenetic tree. Each tip is the most abundant representative sequence for a displayed taxonomic label; genus is shown when available, otherwise the deepest resolved higher rank is displayed. Percentages summarize total final feature-table abundance for that displayed taxonomic label. Internal labels are shown as SH-like local support values when present in the FastTree Newick tree.

- Key signal: Kruskal-Wallis does not show statistically supported alpha-diversity differences (Observed 99% OTUs $p=0.737$, Shannon $p=0.957$, Simpson $p=0.957$, Faith PD $p=0.561$); visual differences should be treated as descriptive.
- Risk: with small cohorts or uneven depth, alpha differences can appear visually strong but remain statistically uncertain.
- Action: use rarefaction plus Kruskal-Wallis together, and treat alpha claims as exploratory when formal p-values are not significant.

Beta-diversity

Table 6. Beta-diversity statistical summary for the final feature table. Bray-Curtis PERMANOVA/PERMDISP are computed with vegan; UniFrac PERMANOVA is computed from exported distance matrices, and UniFrac PERMDISP is computed from distances to group centroids represented across all positive PCoA axes, not only the displayed PC1/PC2 axes.

Metric	PERMANOVA pseudo-F	PERMANOVA p	PERMDISP F	PERMDISP p	Permutations
Bray-Curtis	1.000	1	0.000	1	999
Unweighted UniFrac	1.025	0.43	1.858	0.24	999
Weighted UniFrac	1.521	0.10	5.792	0.03	999

Table 7. Bray-Curtis verification diagnostics for the final 99% OTU feature table. The exact PC1/PC2 percentages and p-values are retained because the exported distance matrix, metadata order, pairwise distances, vegan PERMANOVA, and vegan PERMDISP outputs were checked directly.

Diagnostic	Value
Sample order matches metadata	yes
Metadata missing samples	NA
Group order	Active,Active,Active,Latent,Latent,Latent,Control,Control,Control
Pairwise distances checked	36
Nonzero pairwise distances	36
Pairwise distance min/median/max	1.000/1.000/1.000
Unique rounded distances	1
Positive PCoA eigenvalues	8
PC1 / PC2 variance	12.5% / 12.5%
vegan::adonis2 pseudo-F / p	1.000 / 1.000
vegan::betadisper F / p	0.000 / 1.000

Bray-Curtis note: all pairwise distances from the 99% OTU table were exactly 1.000, which means no 99% OTUs were shared between samples. In this case, the Bray-Curtis PCoA, PERMANOVA, and PERMDISP values are mathematically valid but not useful for biological group comparison, and p=1.000 should not be read as evidence that groups are similar. EMU species-level and UniFrac results are more informative for this dataset.

The species-level Bray-Curtis PCoA uses EMU species relative abundance, while the Bray-Curtis statistics in this section use the final 99% OTU feature table; these inputs answer different composition questions and can produce different p-values.

Table 8. PERMANOVA tests whether predefined groups explain community composition differences using the Bray-Curtis distance matrix. Pseudo-F summarizes the strength of group separation and the p-value is calculated by permutation. Results are treated as statistically supported at $p < 0.05$; results with $p \geq 0.05$ are treated as not statistically supported.

Group Column	Pseudo-F	P-value	Permutations
group	1.000	1.000	999

Table 9. PERMDISP tests whether groups have similar within-group spread using the Bray-Curtis distance matrix. PERMANOVA is easier to interpret when PERMDISP is not significant ($p \geq 0.05$). If PERMDISP $p < 0.05$, apparent group separation may reflect unequal dispersion rather than centroid separation alone.

Group Column	F Value	P-value	Permutations
group	0.000	1.000	999

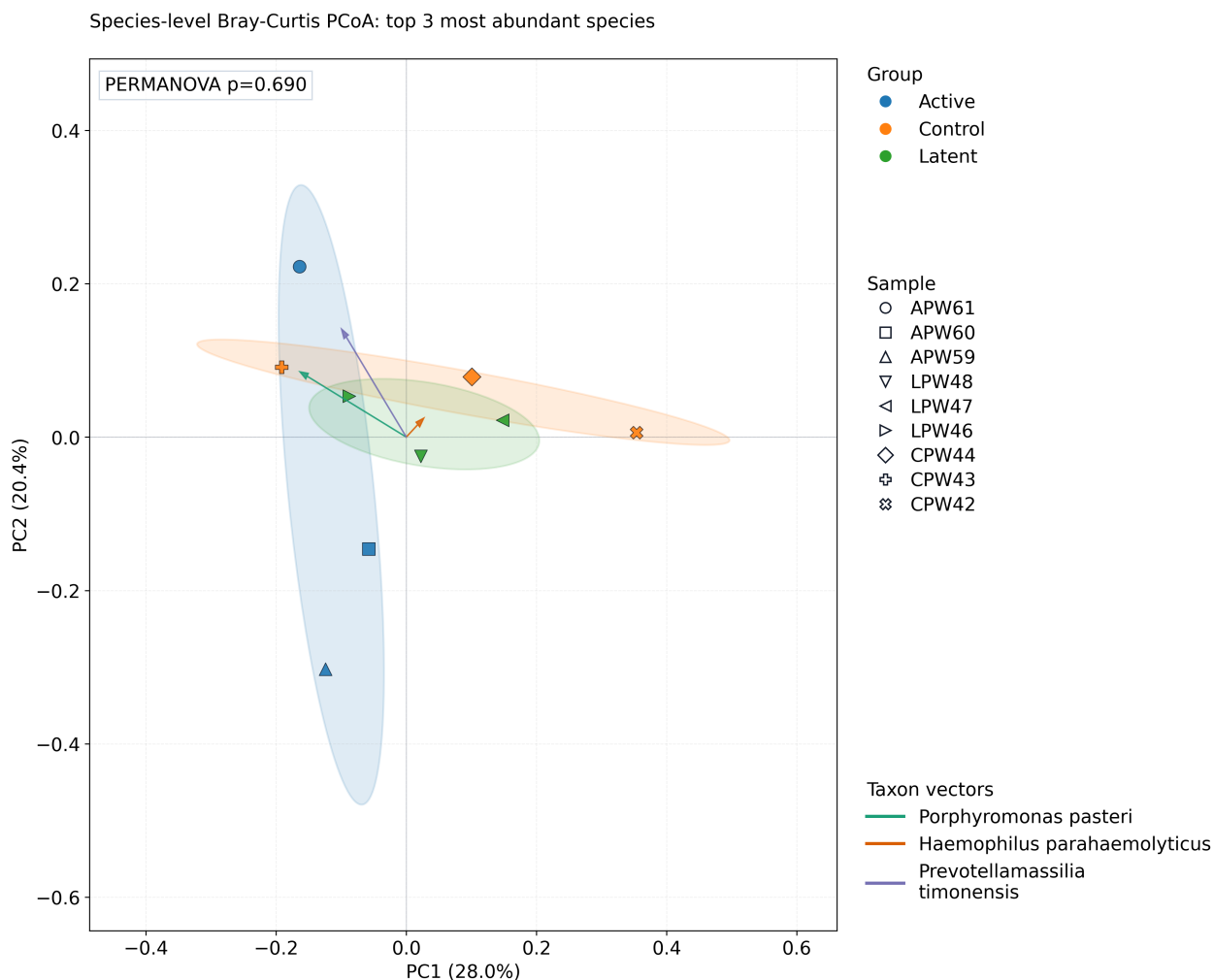


Figure 12. Species-level Bray-Curtis PCoA using EMU species relative abundance. Points are samples colored by group; arrows show descriptive Pearson correlations between the top mean-abundance reportable species and PC1/PC2. This is a beta-diversity ordination of read-level species composition, so it is interpreted with the other between-sample distance plots.

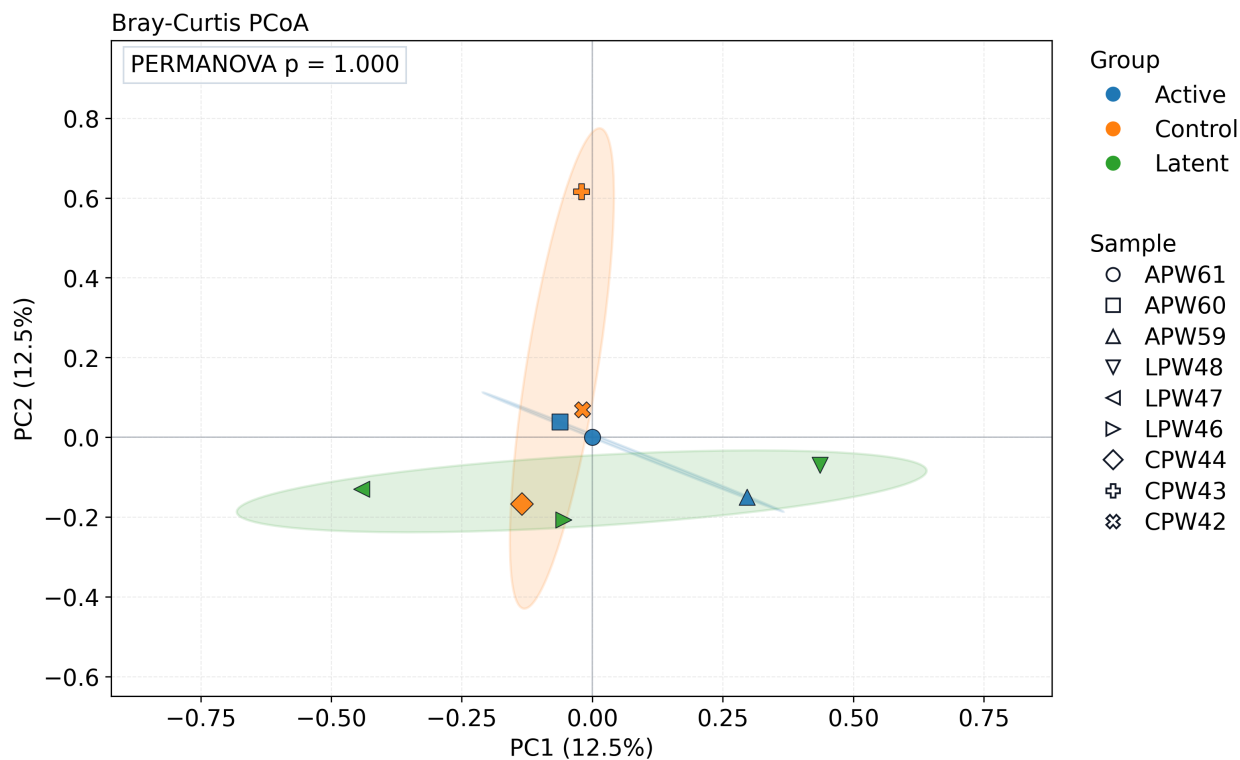


Figure 13. Non-informative 99% OTU Bray-Curtis PCoA plot. All pairwise Bray-Curtis distances are 1.000, so this ordination is retained only as a diagnostic check and should not be used as biological evidence of group similarity or separation. Use EMU species-level Bray-Curtis and UniFrac plots as the primary beta-diversity evidence for this dataset.

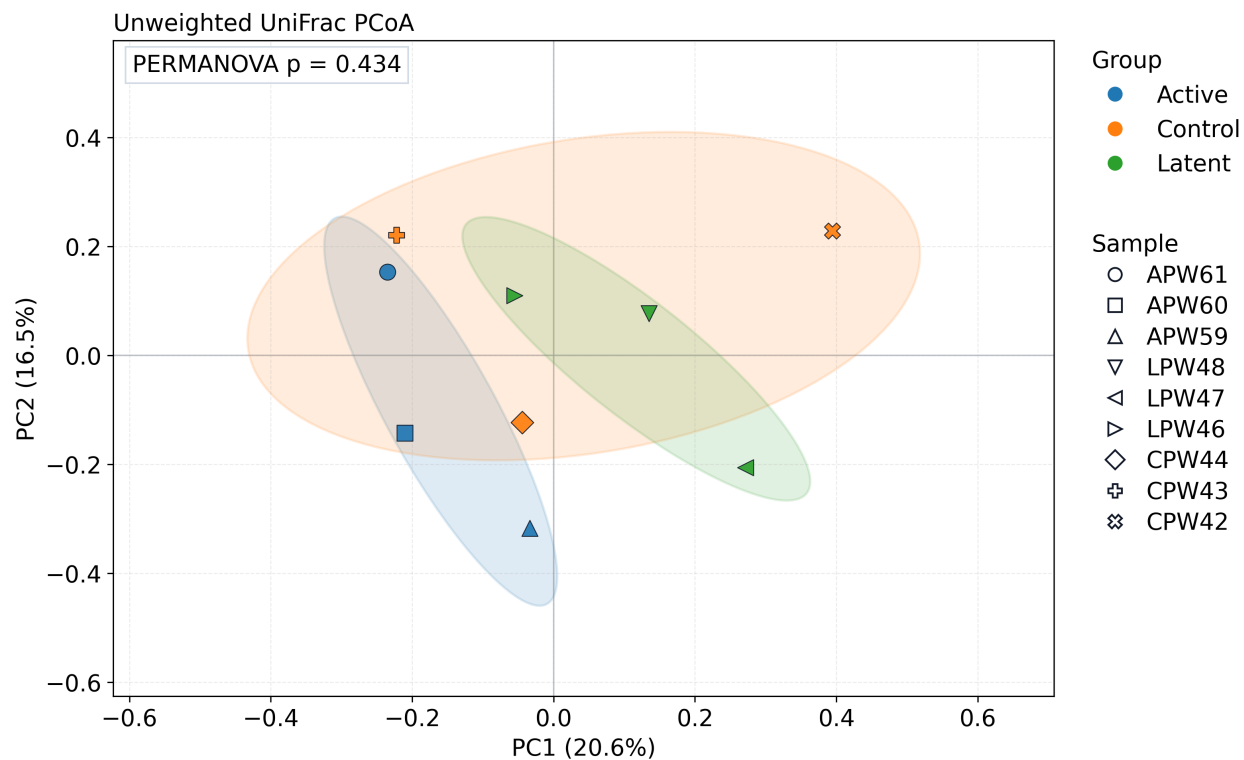


Figure 14. Unweighted UniFrac PCoA uses the rooted phylogenetic tree and presence/absence branch information, so it emphasizes lineage turnover between samples.

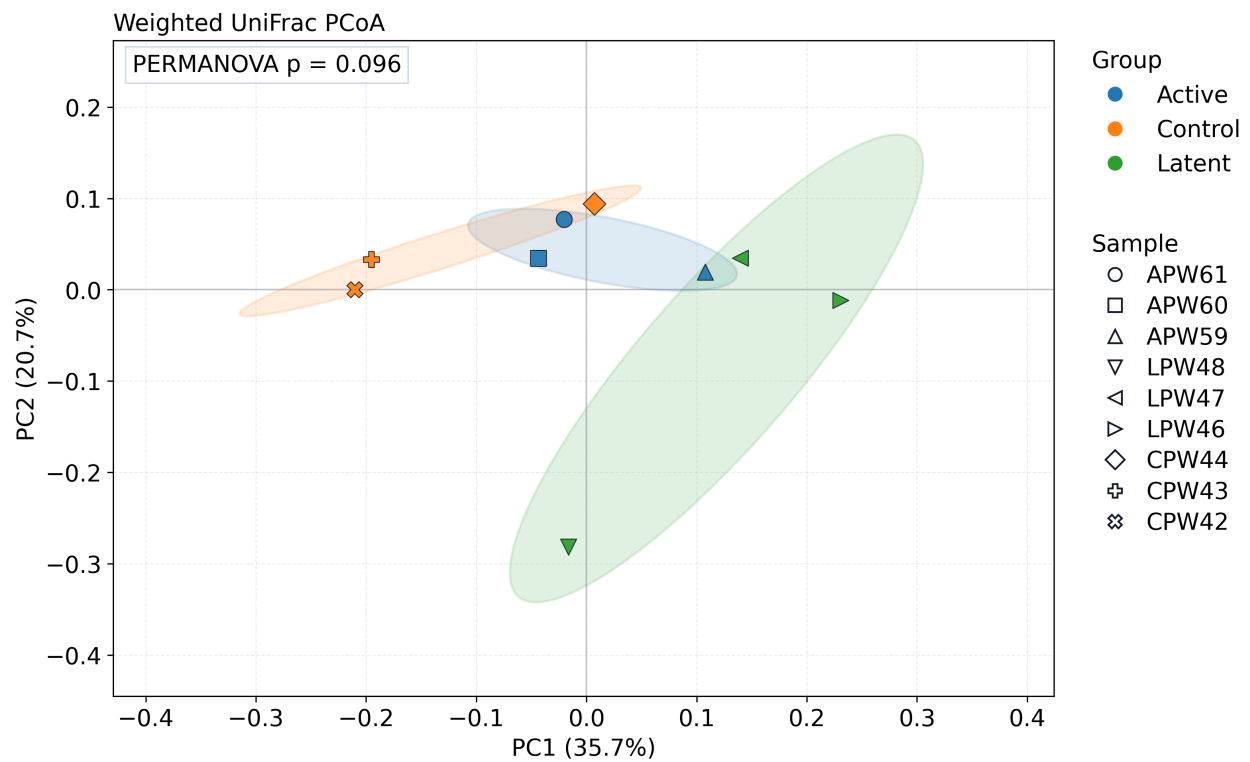


Figure 15. Weighted UniFrac PCoA uses the rooted phylogenetic tree and relative abundance, so it emphasizes abundant phylogenetic lineages.

- Key signal: Bray-Curtis from the 99% OTU table was non-informative because all pairwise distances were 1.000; interpret community composition mainly from EMU species-level profiles and UniFrac metrics (Bray-Curtis $p=1.000$, unweighted UniFrac $p=0.434$, weighted UniFrac $p=0.096$).
- Risk: at least one PERMDISP test was significant (Bray-Curtis $p=1.000$, unweighted UniFrac $p=0.244$, weighted UniFrac $p=0.030$), so PERMANOVA results should be interpreted cautiously. Group separation may reflect unequal within-group dispersion.
- Feature overlap note: a non-significant PERMANOVA result ($p \geq 0.05$) does not mean features are absent or unshared; it means group labels do not explain the overall distance pattern.
- Action: report beta-diversity as inferential only when PERMANOVA and PERMDISP are both interpretable and sample size is adequate; otherwise frame as exploratory.

6 - Predicted Functional Profile

Functional prediction interpretation: PICRUSt2 results in this report are predicted functional potential inferred from bacterial 16S marker-gene representative sequences, not direct metagenomic measurements. To avoid overstating feature-level precision, the workflow uses a separate singleton-filtered, 97% clustered PICRUSt2 input that excludes singleton features and collapses near-identical Nanopore sequence variants before functional inference. Taxonomic composition, Shannon/Simpson alpha diversity, Bray-Curtis beta diversity, and all taxonomy tables remain based on the primary feature table. The clustered PICRUSt2 input is used only for EC, KO, and MetaCyc pathway prediction in this report. Functional heatmaps are normalized to CPM within each sample, log transformed, and row scaled within each function, so displayed color patterns emphasize relative predicted functional profile differences rather than sequencing depth. Functional patterns should be discussed as descriptive predicted community functional potential and qualified by NSTI support.

Table 10. Core PICRUSt2 output volumes retained for interpreting predicted community functional potential. EC (Enzyme Commission) features describe predicted enzyme-level activities, KO features describe predicted KEGG Orthology groups, and MetaCyc pathways summarize higher-level metabolic routes inferred from predicted enzymes. NSTI summarizes how closely the 16S marker sequences match available reference genomes; lower NSTI generally supports higher-confidence prediction. These values are predictions from 16S data, not measured genes.

Metric	Reported value
Predicted EC Features	2108
Predicted KO Features	6066
Predicted MetaCyc Pathways	394
NSTI Mean	0.094
NSTI Median	0.090

Table 11. ALDEx2 differential abundance summary across EC, KO, and MetaCyc pathway feature classes. Each row is one pairwise group contrast. Significant features use the pipeline FDR threshold (Welch test adjusted p-value ≤ 0.05), and the significant fraction reports the percentage of detected features for that contrast.

Feature Class	Contrast	Tested Features	Significant Features	Significant Fraction (%)	Min FDR
EC	Active_vs_Control	1889	0	0	0.99
EC	Active_vs_Latent	1978	0	0	0.98

Feature Class	Contrast	Tested Features	Significant Features	Significant Fraction (%)	Min FDR
EC	Latent_vs_Control	2023	0	0	0.99
KO	Active_vs_Control	5388	0	0	0.99
KO	Active_vs_Latent	5675	0	0	0.97
KO	Latent_vs_Control	5800	0	0	0.98
MetaCyc Pathway	Active_vs_Control	365	0	0	0.97
MetaCyc Pathway	Active_vs_Latent	371	0	0	0.75
MetaCyc Pathway	Latent_vs_Control	385	0	0	0.83

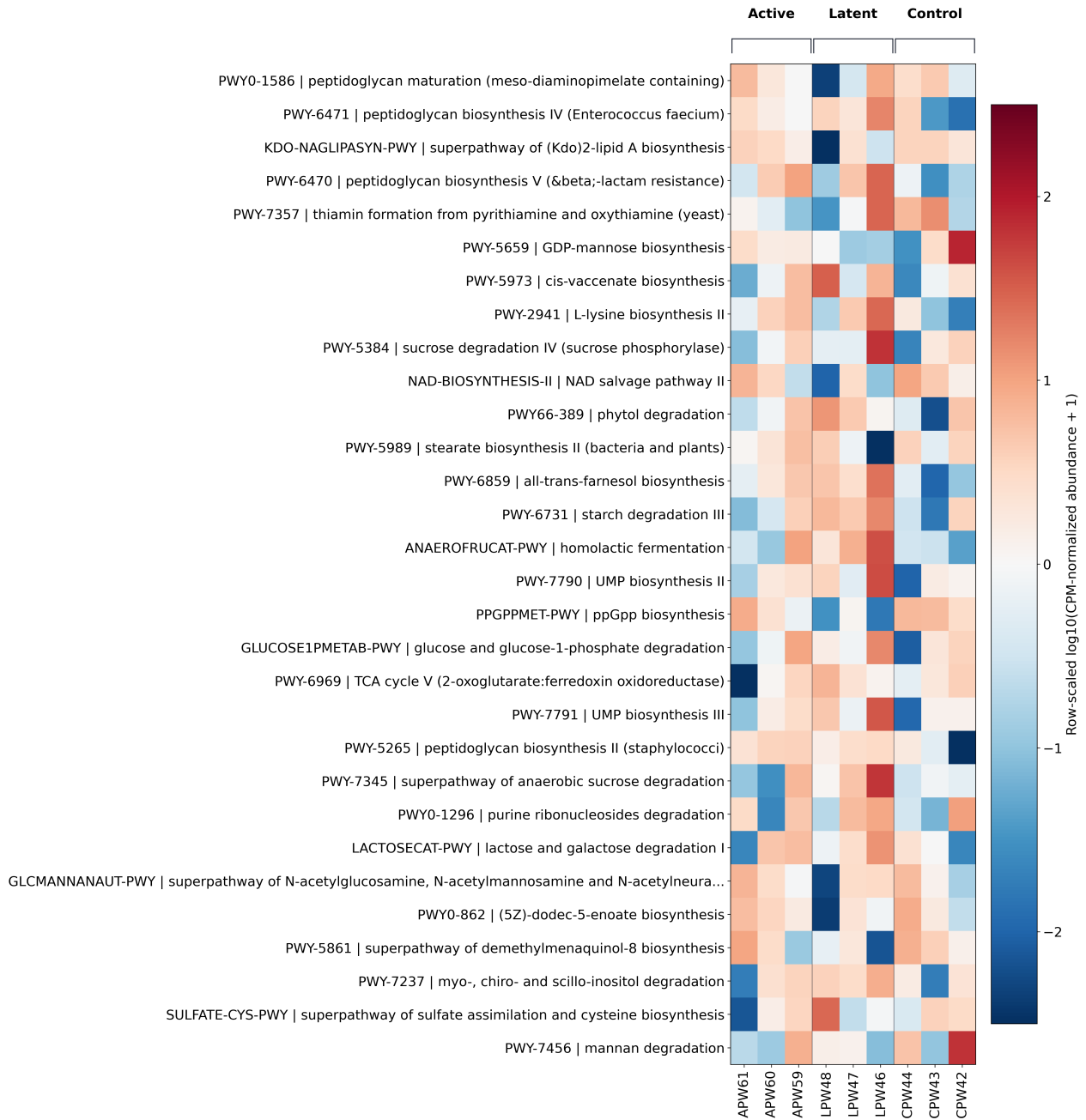


Figure 16. Heatmap of high-variance predicted MetaCyc pathways across samples. Rows are pathways and columns are samples; color intensity shows row-scaled $\log_{10}(\text{CPM-normalized predicted abundance} + 1)$. MetaCyc pathway names do not demonstrate the presence of organisms named in pathway labels.

Top 30 variable predicted EC features

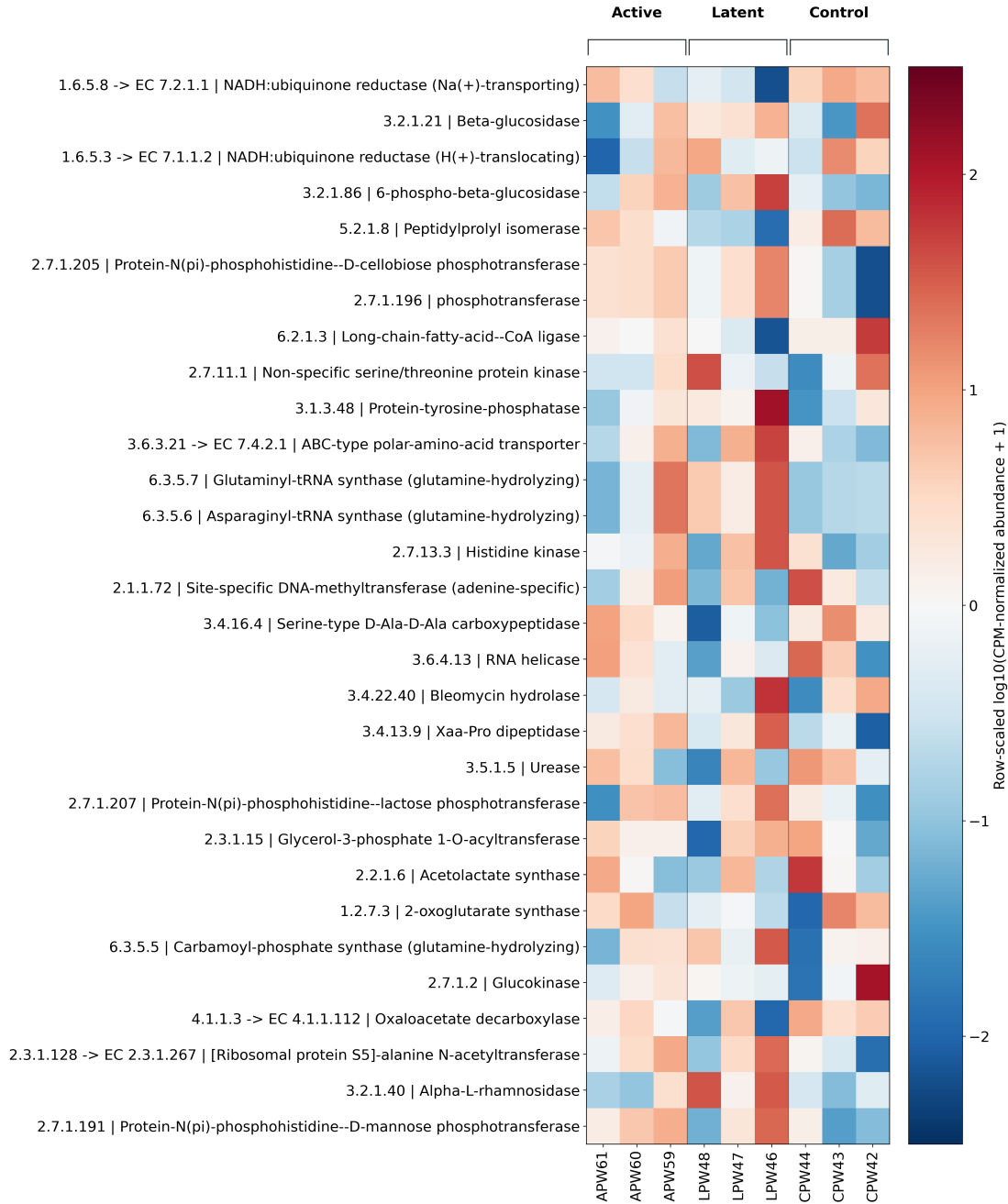


Figure 17. Heatmap of high-variance predicted EC features across samples. Rows are EC-linked enzyme functions and columns are samples; color intensity shows row-scaled $\log_{10}(\text{CPM-normalized predicted abundance} + 1)$, not measured gene abundance or activity.

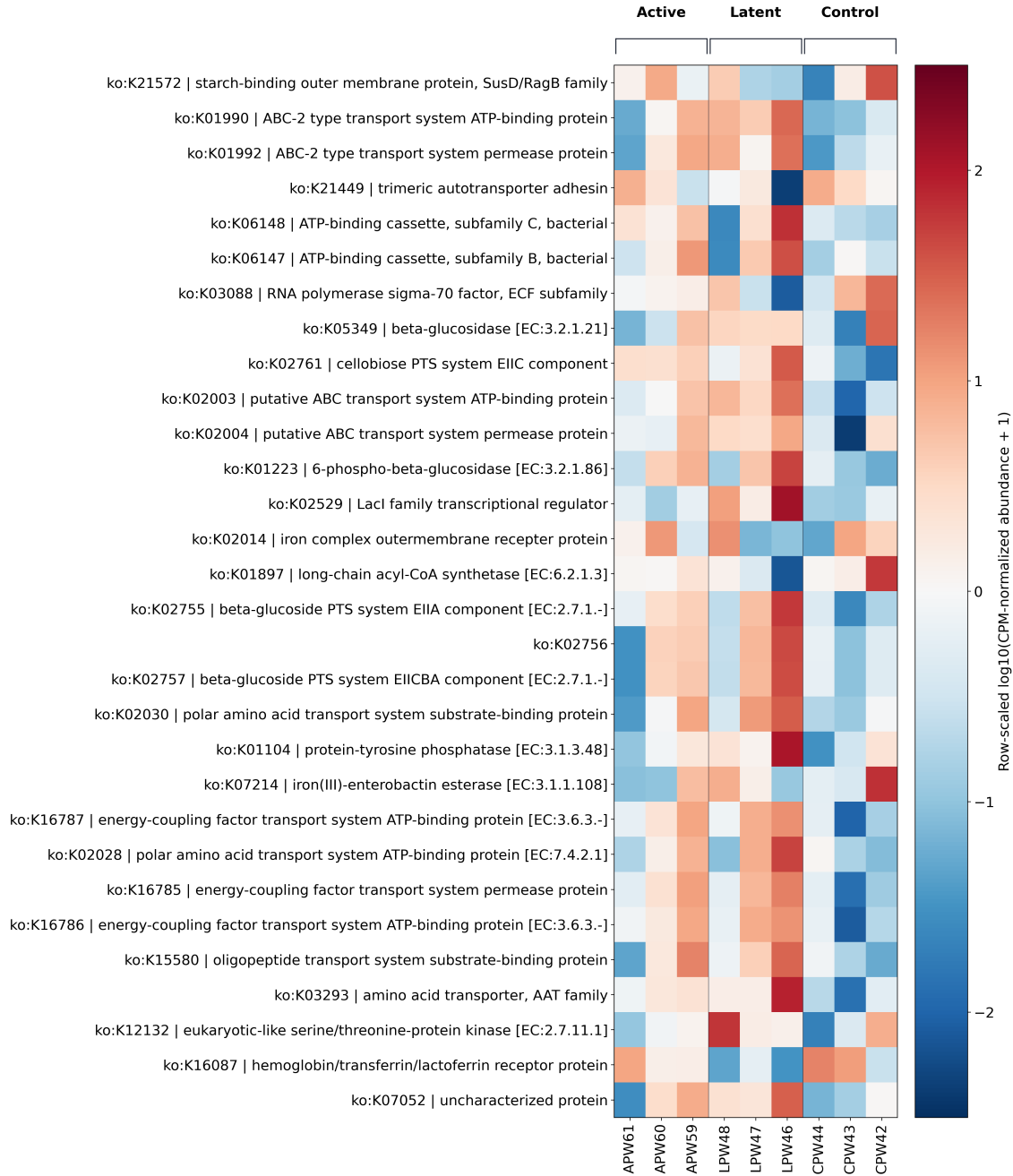


Figure 18. Heatmap of high-variance predicted KO features across samples. Rows are KEGG Orthology features and columns are samples; color intensity shows row-scaled $\log_{10}(\text{CPM-normalized predicted abundance} + 1)$, not measured gene abundance or activity.

ALDEx2 differential abundance volcano plots

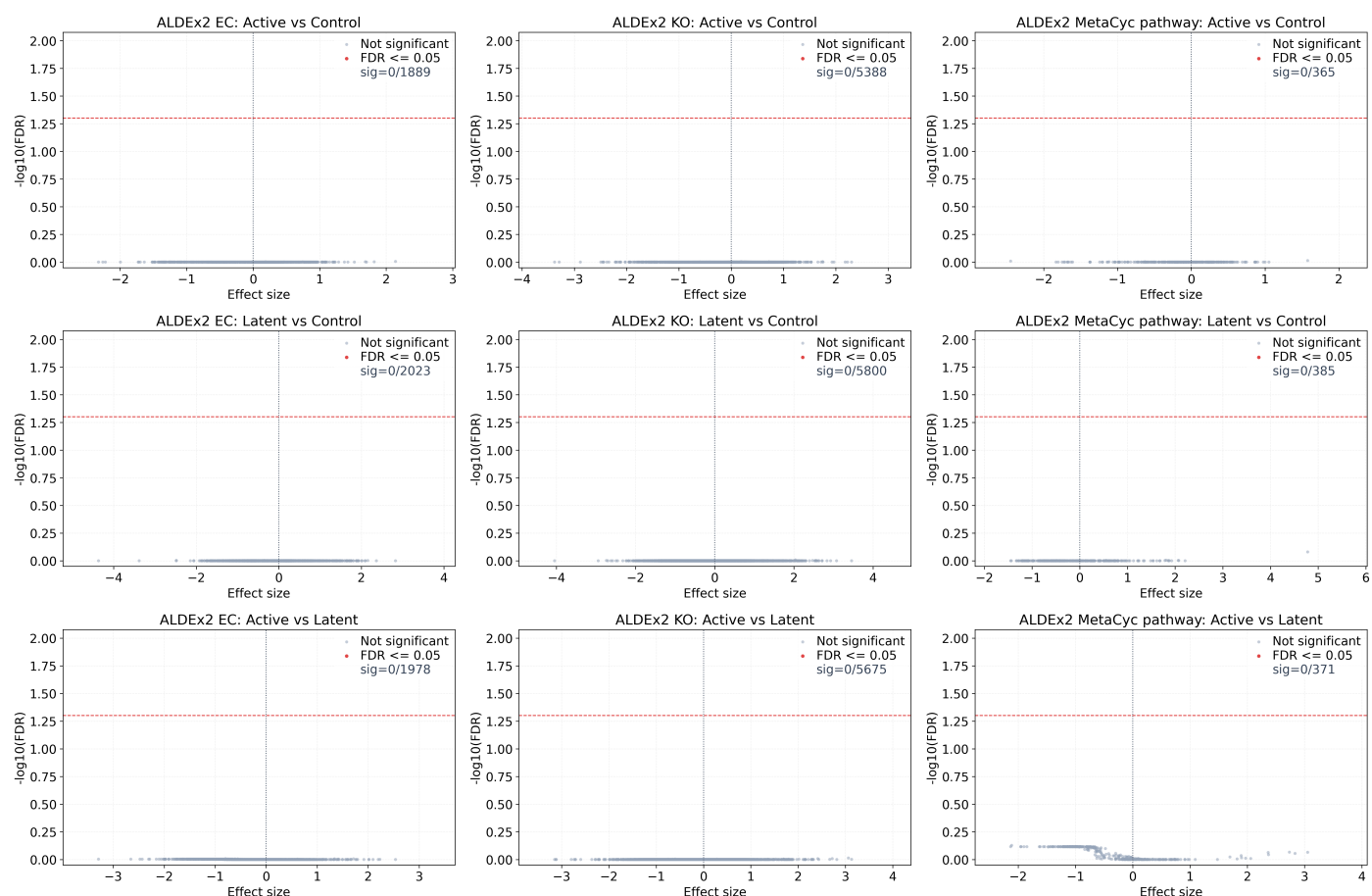


Figure 19. ALDEx2 volcano plots for EC, KO, and MetaCyc pathway features, with each panel showing one pairwise group contrast (effect size vs $-\log_{10}$ FDR). Points farther from zero effect and above the significance threshold represent features with stronger evidence of differential abundance.

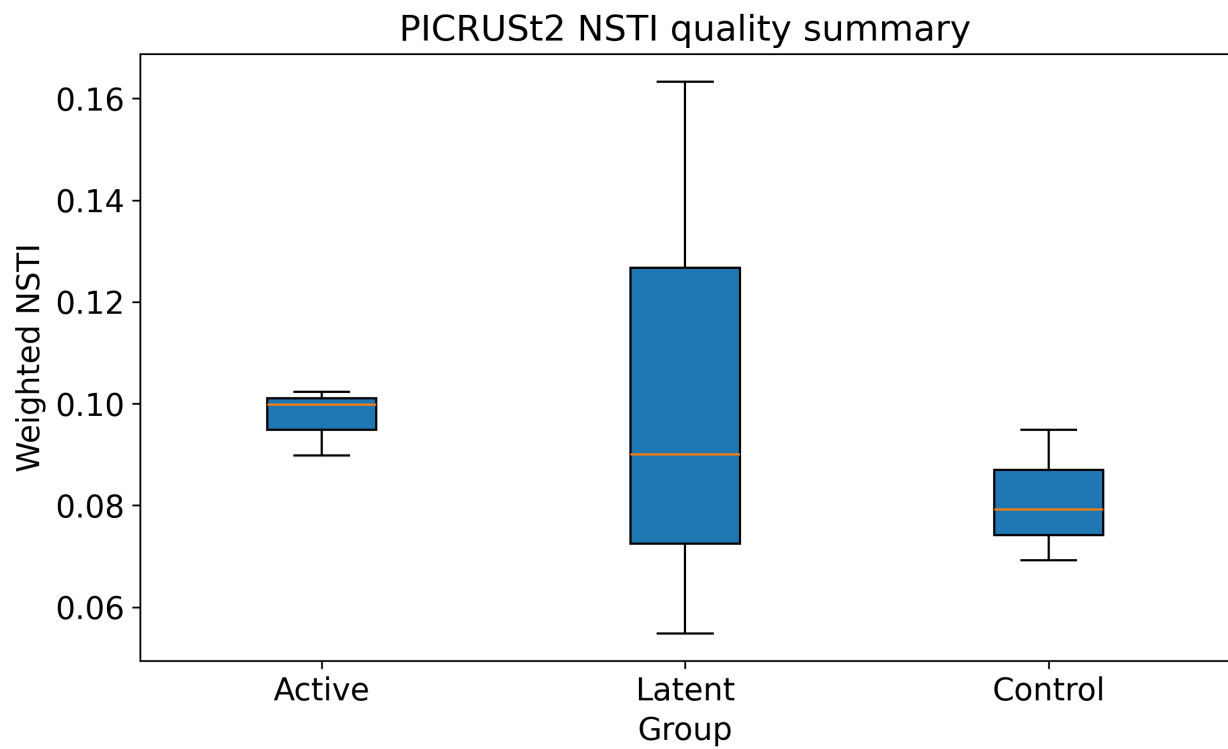


Figure 20. Weighted NSTI distribution summarizing confidence of PICRUSt2 inference for each sample/group. Lower NSTI values indicate closer reference-genome representation and generally higher confidence in predicted functional profiles.

NSTI confidence summary: $n=9$, median=0.090, range=0.055-0.163. Lower values indicate stronger reference support for functional prediction.

- Key signal: no ALDEx2 EC, KO, MetaCyc pathway features passed the report significance threshold; predicted functional differences should be treated as descriptive.
- Risk: PICRUSt2 outputs are inferred from 16S and clustered only for functional prediction, so claims can be overstated when framed as direct gene measurements or directly measured functions.
- Action: keep functional language explicitly predictive, and prioritize claims supported by NSTI quality, multi-feature agreement, and corrected differential testing.

7 - Reproducibility and Artifacts Module

Table 12. This table summarizes the output package layout.

Artifact	Path
Final PDF report	output/report.pdf
QC summaries and MultiQC report	output/o1_qc/
QIIME 2 SILVA taxonomy tables	output/o2_taxonomy_qiime2_silva/
EMU species reportable calls and abundance	output/o3_taxonomy_emu_species/
Diversity and group statistics	output/o4_diversity/
PICRUSt2 functional prediction summaries	output/o5_functional_prediction_picrust2/
Figures	output/o6_figures/
Methods, software versions, citations, and parameters	output/o7_run_metadata/

8 - Citations and References

Table 13. Core software, database, and workflow citations used by the analysis are listed here instead of being shipped as a separate citations TSV.

Tool	Citation
NanoPlot	De Coster et al. 2018. NanoPack: visualizing and processing long-read sequencing data. <i>Bioinformatics</i> 34(15):2666-2669.
cutadapt	Martin 2011. Cutadapt removes adapter sequences from high-throughput sequencing reads. <i>EMBnet.journal</i> 17(1):10-12.
chopper	De Coster W. chopper: a Rust command-line tool for filtering and trimming long reads. Version recorded in this workflow; source code and documentation: https://github.com/wdecoster/chopper .
vsearch	Rognes et al. 2016. VSEARCH: a versatile open source tool for metagenomics. <i>PeerJ</i> 4:e2584. doi:10.7717/peerj.2584.
NanoPulse	FOI Bioinformatics 2026. NanoPulse: de novo Nanopore amplicon clustering and consensus generation.
QIIME 2	Bolyen et al. 2019. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. <i>Nature Biotechnology</i> 37:852-857. doi:10.1038/s41587-019-0209-9.
SILVA	Quast et al. 2013. The SILVA ribosomal RNA gene database project. <i>Nucleic Acids Research</i> 41(D1):D590-D596. doi:10.1093/nar/gks1219; Yilmaz et al. 2014. The SILVA and All-species Living Tree Project releases. <i>Nucleic Acids Research</i> 42(D1):D643-D648. doi:10.1093/nar/gkt1209.
EMU	Curry et al. 2022. Emu: species-level microbial community profiling of full-length 16S rRNA Oxford Nanopore sequencing data. <i>Nature Methods</i> 19:845-853. doi:10.1038/s41592-022-01520-4.
MAFFT	Katoh and Standley 2013. MAFFT multiple sequence alignment software version 7. <i>Molecular Biology and Evolution</i> 30(4):772-780. doi:10.1093/molbev/mst010.
FastTree	Price et al. 2010. FastTree 2: approximately maximum-likelihood trees for large alignments. <i>PLoS ONE</i> 5(3):e9490. doi:10.1371/journal.pone.0009490.
UniFrac	Lozupone and Knight 2005. UniFrac: a new phylogenetic method for comparing microbial communities. <i>Applied and Environmental Microbiology</i> 71(12):8228-8235. doi:10.1128/AEM.71.12.8228-8235.2005.
PICRUSt2	Douglas et al. 2020. PICRUSt2 for prediction of metagenome functions. <i>Nature Biotechnology</i> 38:685-688. doi:10.1038/s41587-020-0548-6.
ALDEx2	Fernandes et al. 2014. Unifying the analysis of high-throughput sequencing datasets: characterizing RNA-seq, 16S rRNA gene sequencing and selective growth experiments by compositional data analysis. <i>Microbiome</i> 2:15. doi:10.1186/2049-2618-2-15.
vegan	Oksanen et al. vegan: Community Ecology Package. R package documentation for the installed workflow version.
R	R Core Team. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Tool	Citation
MultiQC	Ewels et al. 2016. MultiQC: summarize analysis results for multiple tools and samples in a single report. <i>Bioinformatics</i> 32(19):3047-3048. doi:10.1093/bioinformatics/btw354.